

SYDNEY GRAMMAR SCHOOL



2016

FORM V
EXAMINATION

Physics

Working Time: 2 hours

1:00 pm Monday 23 MAY

General Instructions

- Working time – 2 hours
- Board-approved calculators may be used
- Write using blue or black pen
- Draw diagrams using pencil
- A Data Sheet, Formula Sheet and Periodic Table are provided at the back of this paper
- Write your name at the top of the Multiple Choice Answer Sheet & all pages of Part B.
- Hand in your Multiple Choice Sheet and all of Part B in one bundle. (Do not staple together)

Total marks (99)

This paper has two parts: Part A and Part B

Part A

Total marks (13)

- Attempt ALL Questions
- Allow about 15 minutes for this Part

Part B

Total marks (86)

- Attempt ALL Questions
- Allow about 1 hour 45 minute for this Part

CHECKLIST

Each boy should have the following:

1 Question Paper

1 Multiple Choice Answer Sheet

1 – AAH	2 – SRW	3 – PCK	
4 – DGB	5 – MRW	6 – SRW	7 – MTK

EXAMINERS: MRW/AAH/PCK/SRW/MTK

Part A**Total marks (13)****Attempt ALL Questions****Allow about 15 minutes for this Part**

Use the multiple-choice Answer Sheet.

Select the alternative A, B, C or D that best answers the question. Fill the response circle completely.

Sample

$2 + 4 =$

(A) 2

(B) 6

(C) 8

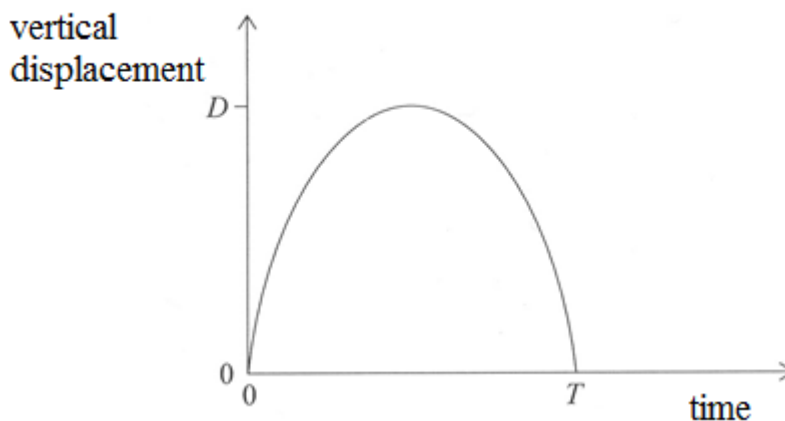
(D) 9

☐ A☒ B☐ C☐ D

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

☒ A☒ B☐ C☐ DIf you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.☒ A☒ B*correct*☐ C☐ D

- 1 Which of the following lists contains three vector quantities?
- (A) Velocity, mass, acceleration
(B) Energy, velocity, time
(C) Work, displacement, velocity
(D) Acceleration, momentum, force
- 2 Which of the following is **always** true for an object travelling in a straight line at a constant speed?
- (A) The friction is zero.
(B) No forces are acting on it.
(C) The momentum of the object is zero.
(D) No net force acts on the object.
- 3 A ball is thrown vertically upwards from the ground.
The graph below shows the variation of the vertical displacement of the ball with time.

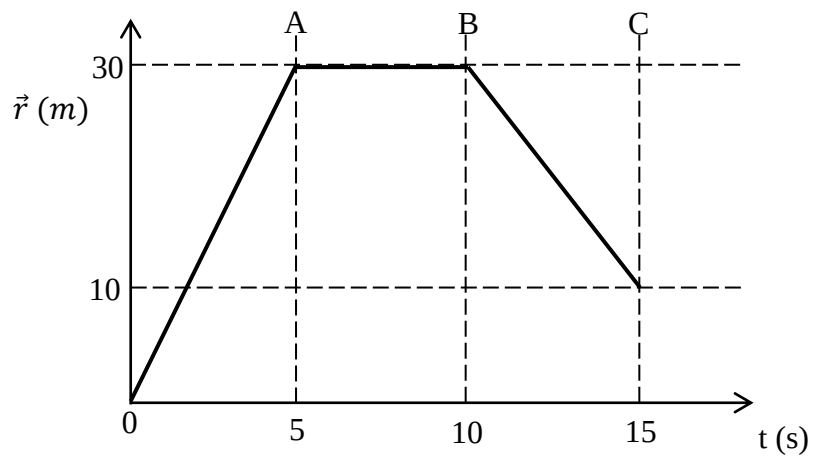


Determine which of the following is the final vertical displacement of the ball at time T and the average speed between time $t = 0$ and time T ?

	Final vertical Displacement	Average Speed
(A)	0	0
(B)	0	$2D/T$
(C)	D	$2D/T$
(D)	$2D$	0

The following information applies to Questions 4 and 5:

The displacement versus time graph below shows the motion of an object along a straight line.

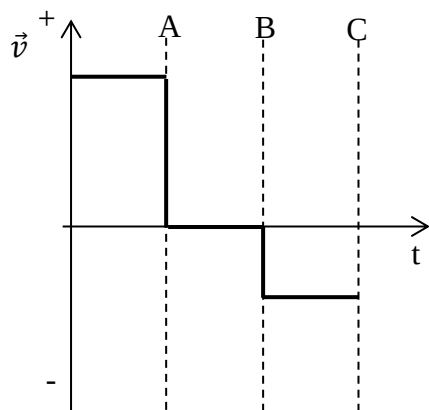


4 What is the total distance travelled by the object?

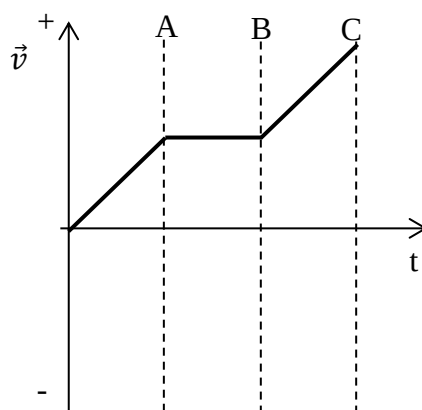
- (A) 10 m
- (B) 20 m
- (C) 40 m
- (D) 50 m

- 5 Which of the following graphs best represents the velocity versus time graph for the motion described in the displacement versus time graph on the previous page?

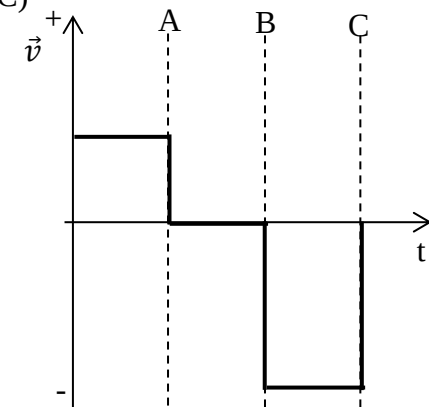
(A)



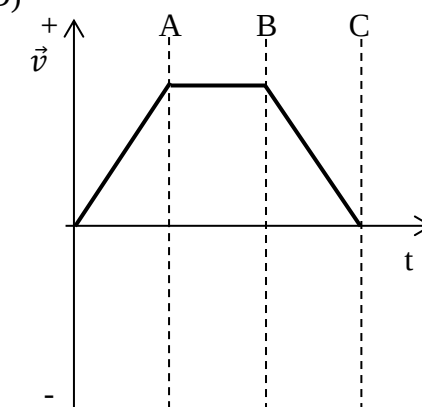
(B)



(C)



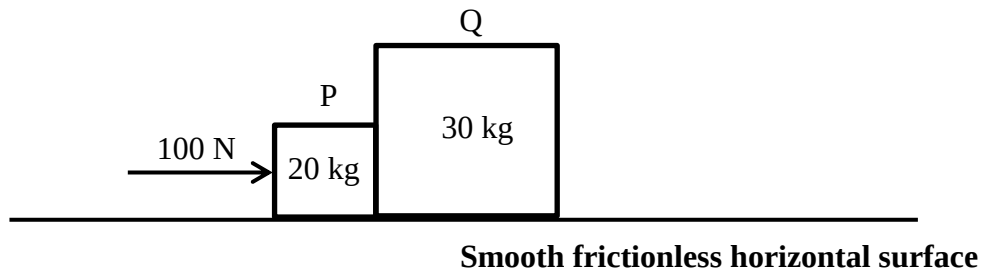
(D)



- 6 A 40 N force is applied horizontally to a 10 kg block which is initially at rest. After 5.0 seconds, the velocity of the block is 12 m s^{-1} . What is the magnitude of the frictional force acting on the block?

- (A) 0 N
 (B) 16 N
 (C) 24 N
 (D) 40 N

- 7 Consider the following arrangement of blocks, labelled P and Q.

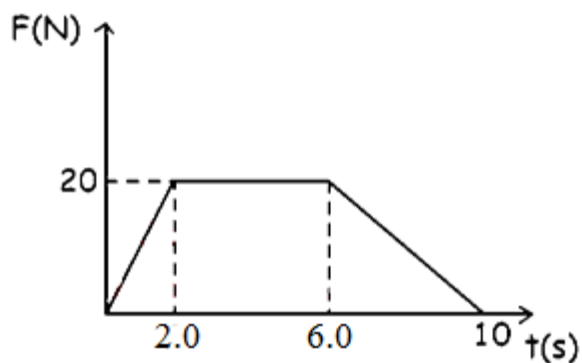


A constant, horizontal force of 100 N is applied to block P.

Determine the magnitude of the force block P exerts on block Q.

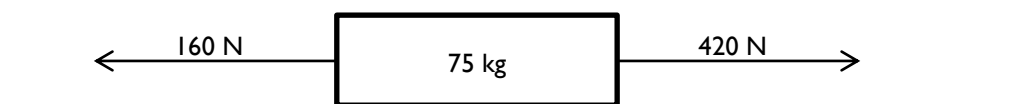
- (A) 40 N
 - (B) 60 N
 - (C) 80 N
 - (D) 100 N
- 8 A block moves at a speed of 10 m s^{-1} across a frictionless, horizontal surface. No net force is acting on this block. What is the magnitude of the velocity of the block 10 seconds later?
- (A) 0 m s^{-1}
 - (B) Less than 10 m s^{-1}
 - (C) 10 m s^{-1}
 - (D) Greater than 10 m s^{-1}

- 9 The following graph shows how the net force acting on a toy car varies with time.



What is the magnitude of the total change in momentum of the toy car at the end of 10 seconds?

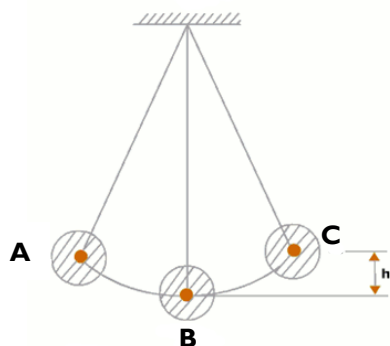
- (A) 240 kg m s^{-1}
(B) 200 kg m s^{-1}
(C) 140 kg m s^{-1}
(D) 100 kg m s^{-1}
- 10 Consider the block shown below which is initially at rest on a frictionless horizontal surface and then acted upon by two forces simultaneously.



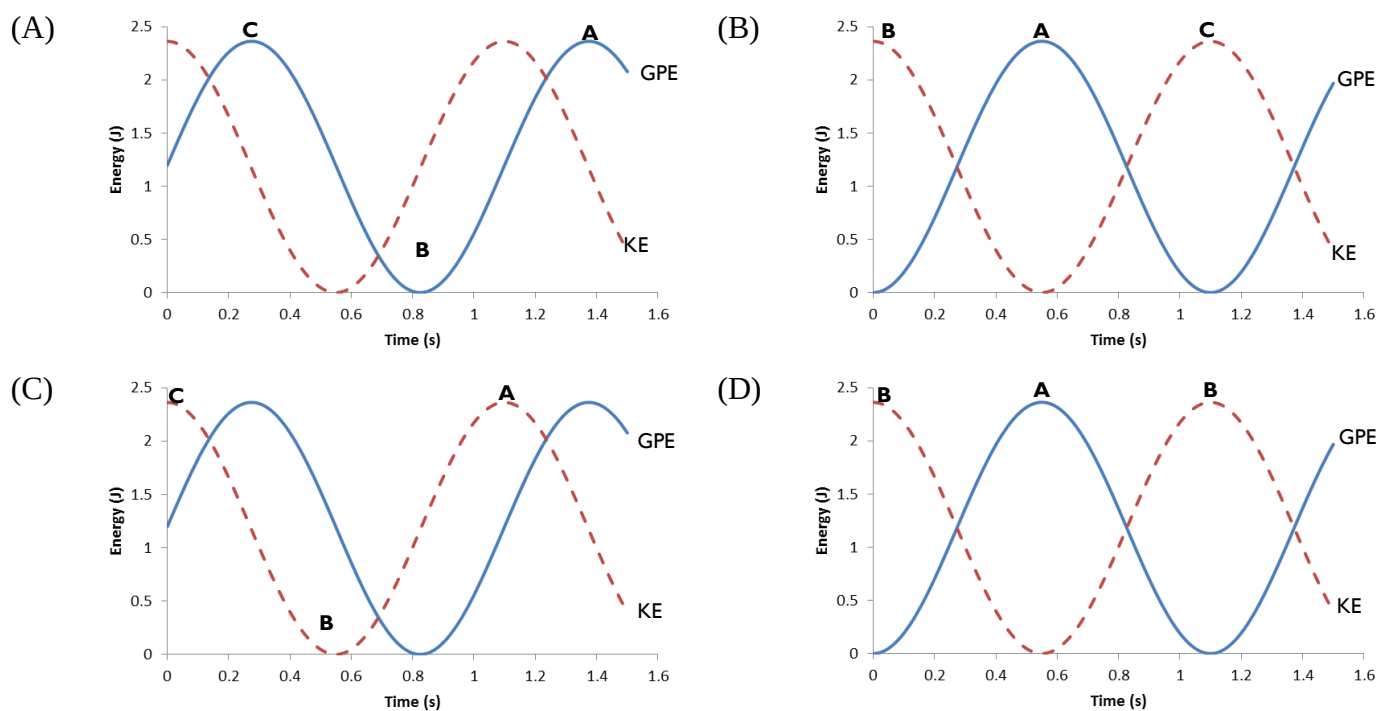
Determine the kinetic energy of the block when it has moved 5 metres.

- (A) 3.5 J
(B) 260 J
(C) 1300 J
(D) 2100 J

- 11 A pendulum swings backwards and forwards through positions A, B and C.



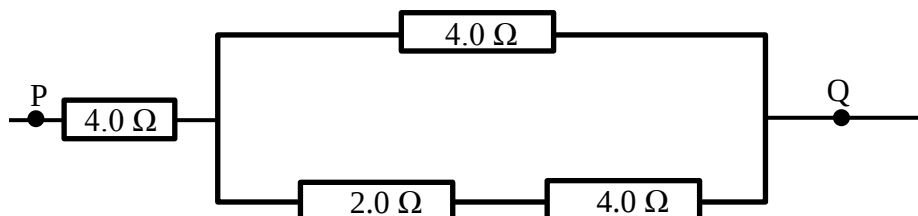
Which of the following graphs **best** represents how the gravitational potential energy (GPE) and kinetic energy (KE) vary with time?



- 12 A resistor has a steady current of 5.0 A flowing through it in 10 s.
How many electrons pass through the resistor in 10 s?

- (A) 8.0×10^{-18}
 (B) 3.1×10^{20}
 (C) 2.5×10^{-3}
 (D) 1.6×10^{20}

- 13 Consider the following combination of resistors.



What is the total resistance between points P and Q in the above section of an electric circuit?

- (A) $2.0\ \Omega$
- (B) $4.0\ \Omega$
- (C) $6.4\ \Omega$
- (D) $11.6\ \Omega$

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Class

Name

Part B**Total marks (86)****Attempt ALL Questions****Allow about 1 hour and 45 minutes for this Part**

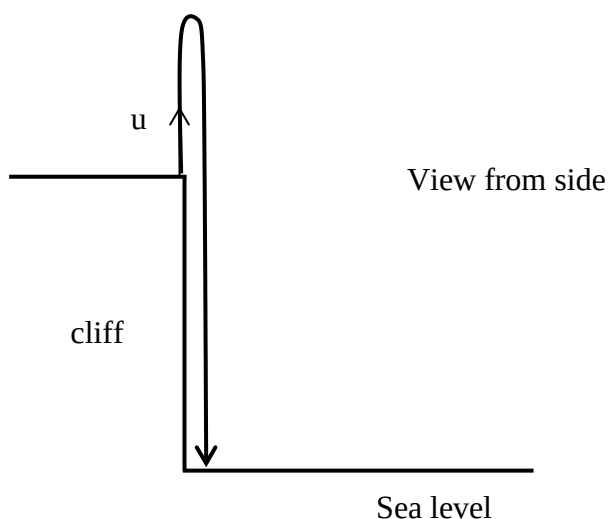
Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 14 (5 marks)**Marks**

A ball was thrown vertically upwards at a speed of 10.0 m s^{-1} from the top of a cliff. The time of flight for the ball to hit the sea at the bottom of the cliff was 6.92 s.

Assume no air resistance acts on the ball.



- (a) Calculate the speed with which the ball hits the sea at the bottom of the cliff.

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2**Question 14 continued on next page.**

Question 14 continued.

Marks

(b) Calculate the height of the cliff top above sea level.

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(c) What is the acceleration of the ball at the highest point?

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Class

Name

Marks

Question 15 (6 marks)

A police motor bike is stationary by the side of a road. An L-plate driver, exceeding the speed limit, passes the police motor bike at a constant speed of 108 km h^{-1} . The police motor bike sets off to catch the L-plater just as he passes. The police motor bike accelerates uniformly at 6.5 m s^{-2} for 6.0 s and then continues at a constant speed.

- (a) Convert the L-plate driver's speed to ms^{-1} .

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- (b) Calculate the speed of the police motor bike at 6 s .

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- (c) Calculate the distance travelled by police motor bike in the first 6 s .

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- (d) Calculate the total time for the police motor bike to draw level with the L-plate driver.

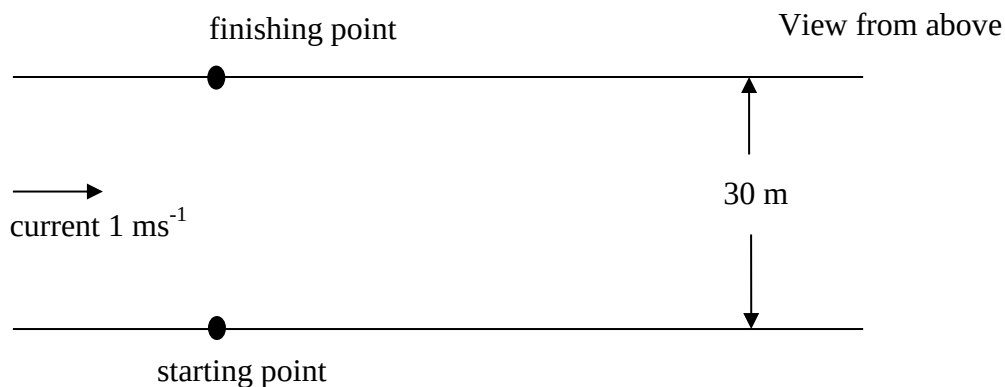
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- (e) Calculate how far the L-plate driver has travelled before the police motor bike catches up.

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Question 16 (3 marks)**Marks**

A swimmer competes in a swimming race across a 30 m wide river which has a constant downstream current of 1.0 ms^{-1} . He swims slightly against the current so that he ends up exactly opposite the starting point as shown in the following diagram.



The swimmer can swim at 2 ms^{-1} in still water.

By drawing velocity vectors in the space provided below, determine the time it takes the swimmer to swim from the starting point to the finishing point.

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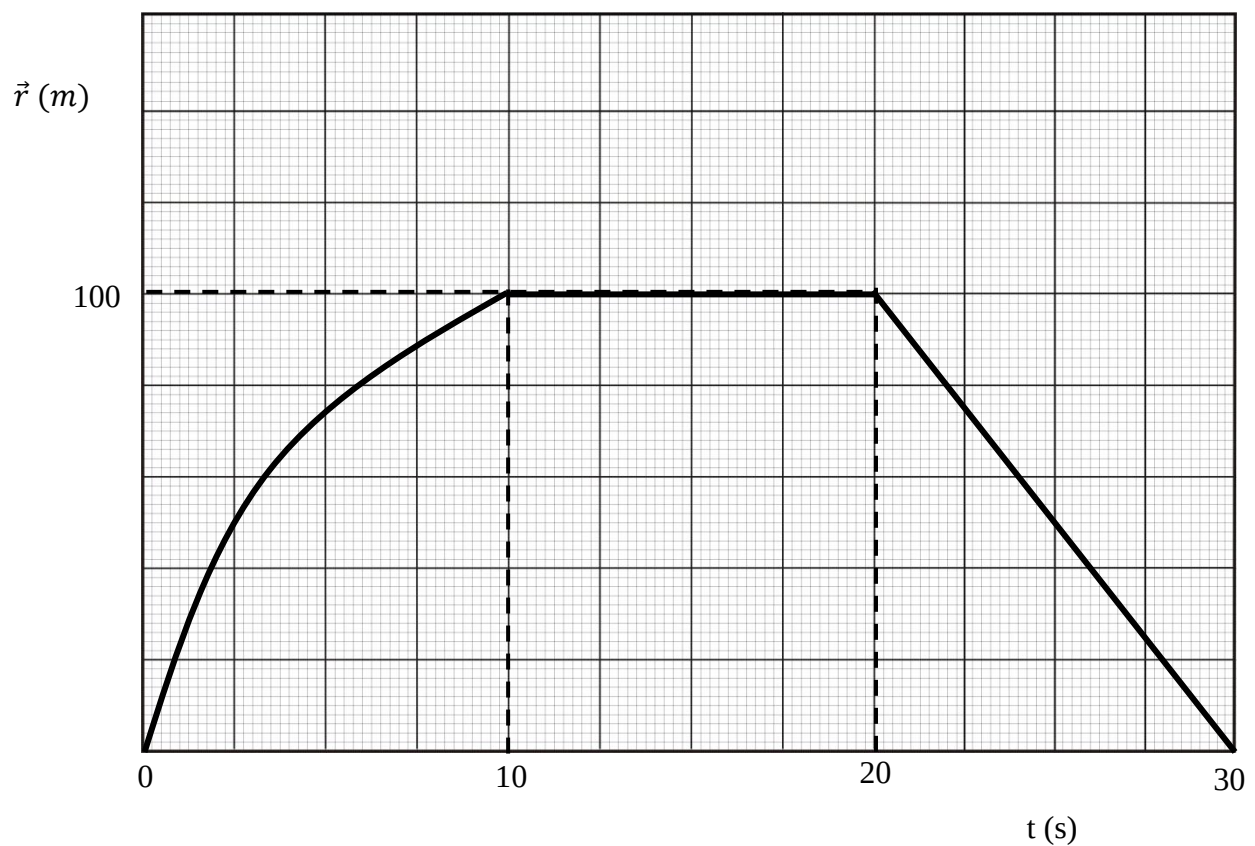
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Class

Name

Question 17 (5 marks)**Marks**

The diagram below shows the displacement versus time graph for a car travelling along a straight road.



From the graph, determine:

- (a) the average speed of the car between 0 and 30 s.

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Question 17 continued on next page.

Question 17 continued.

Marks

(b) the average velocity of the car between 20 and 30 s.

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(c) the instantaneous velocity of the car at 5 s.

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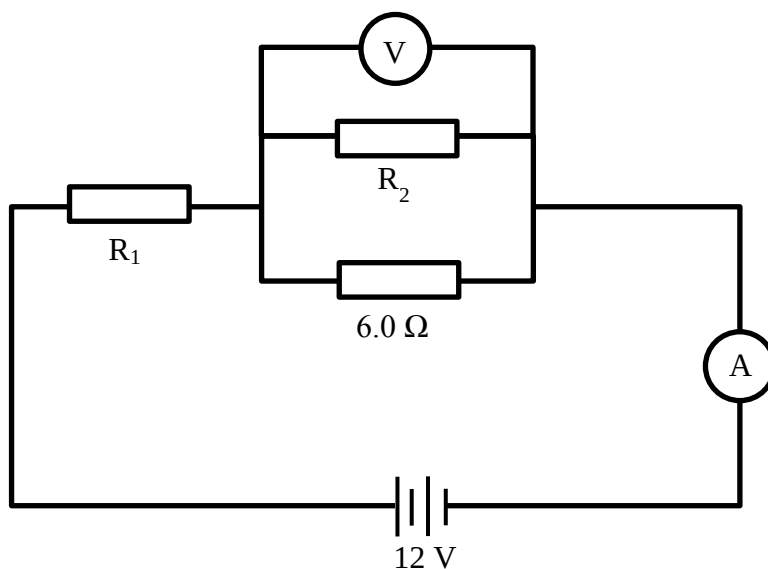
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ClassName**Question 18** (4 marks)**Marks**

This question refers to the electric circuit shown below.



The voltmeter has a reading of 4.0 V and the ammeter has a reading of 3.0 A.

- (a) Calculate the voltage across the unknown resistor, R_1 .

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- (b) Calculate the value of the unknown resistor, R_1 .

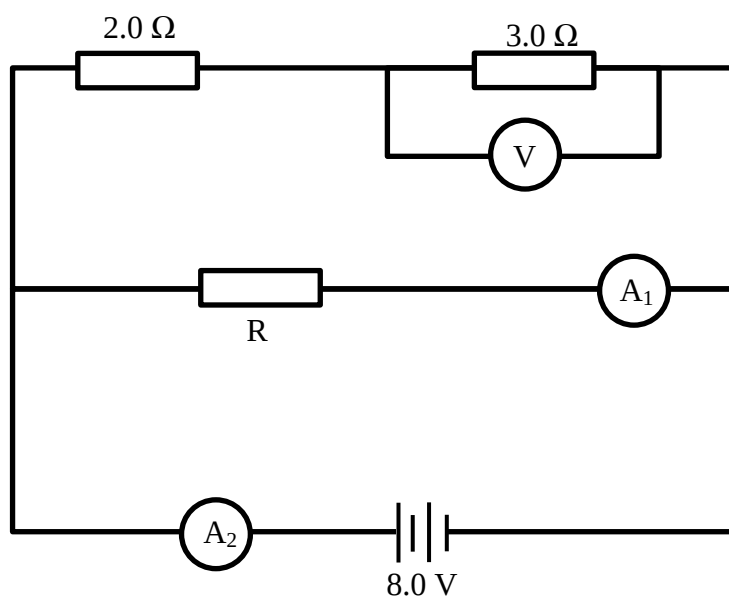
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- (c) Calculate the value of the unknown resistor, R_2 .

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Question 19 (4 marks)**Marks**

This question refers to the electric circuit shown below.



The reading on the ammeter, A_1 is 0.80 A.

- (a) Calculate the value of the unknown resistor, R .

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- (b) Determine the reading on the voltmeter in the circuit above.

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- (c) Calculate the reading on the ammeter A_2 .

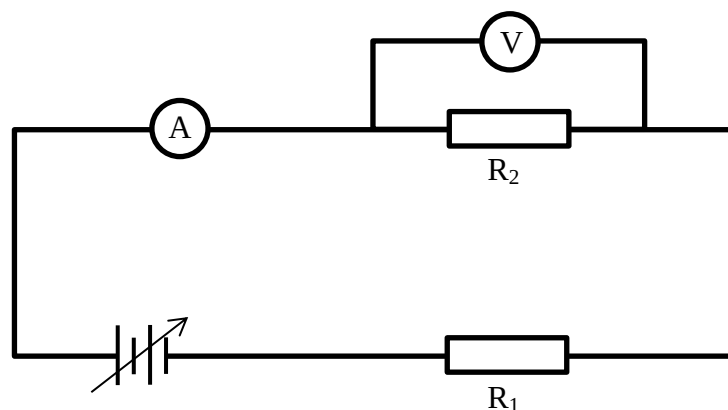
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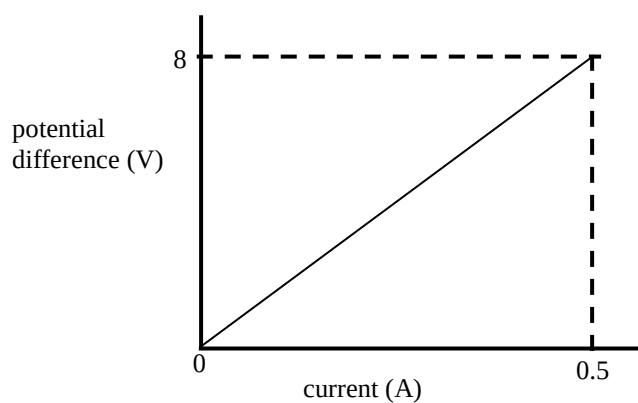
Name

Question 20 (3 marks)**Marks**

The following circuit was set up with two resistors, R_1 and R_2 , and a battery.



The following graph shows how the potential difference across resistor, R_2 , varies as the current through resistor, R_2 , is changed.



- (a) Calculate the value of the resistance of resistor, R_2 .

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Question 20 continued on next page

Question 20 continued.**Marks**

- (b) When the potential difference of the battery is 8.0 V, it supplies a current of 0.40 A to the circuit.

Calculate the value of the unknown resistor, R_1 .

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2**Question 21 (2 marks)**

A 100 kg man stands on a set of bathroom scales in a lift and notices that the scales read 153 kg.

Determine the acceleration of the lift.

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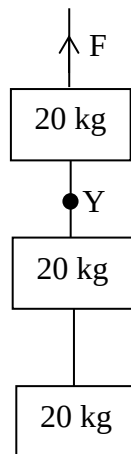
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Class

Name

Question 22 (4 marks)**Marks**

Three 20 kg masses are pulled **upwards** with a constant force F such that the tension force in the rope at point Y is 500 N.



- (a) Determine the magnitude of the acceleration of the masses.

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- (b) Determine the magnitude of the constant force, F .

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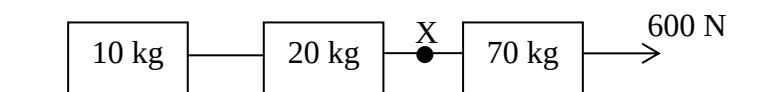
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Question 23 (4 marks)**Marks**

Three masses 10 kg, 20 kg and 70 kg are connected by massless and unstretchable strings and pulled along a frictionless, horizontal surface by a constant horizontal force of 600 N as shown below.



- (a) Calculate the magnitude of the acceleration of the masses.

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- (b) Calculate the tension force in the string at X.

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- (c) The same arrangement of three masses is now placed on a rough surface. If the frictional force which now acts on each mass is equal to 0.2 times the weight of each mass, determine the magnitude of the new acceleration of the system when the constant horizontal force of 600 N is applied.

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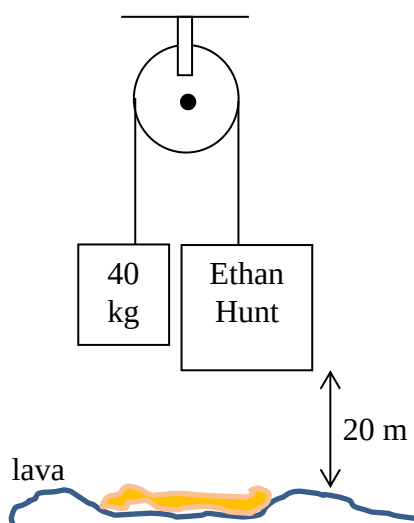
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Class

Name

Question 24 (6 marks)**Marks**

In “*Mission Impossible Twelve*”, Ethan Hunt of mass 80 kg finds himself attached to a 40 kg mass via a pulley above a pool of lava as shown below.



- (a) Assuming the system is released from rest, what time would elapse before Ethan reaches the surface of the lava?

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- (b) Ethan starts to climb up the rope with an acceleration of 1.0 m s^{-2} **relative to the lava**.

- (i) Determine the tension in the rope while Ethan is climbing.

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Question 24 continued on next page.

Question 24 continued.

Marks

- (ii) Determine the magnitude of the acceleration of the 40 kg mass while Ethan climbs the rope.

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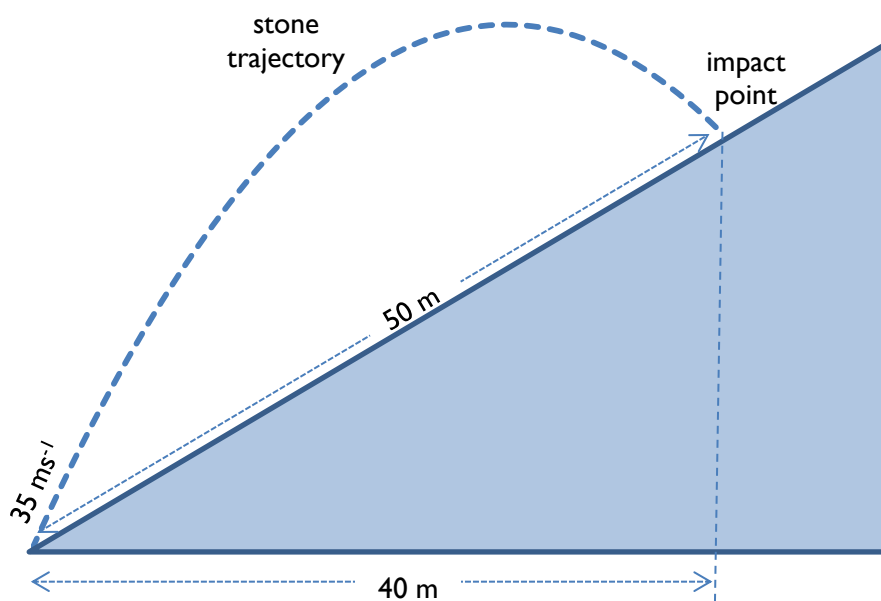
Class

Name

Marks

Question 25 (4 marks)

A stone of mass 0.25 kg is thrown into the air with an initial speed of 35 m s^{-1} and it lands 50 m up a slope as shown in the following diagram.



- (a) Determine the initial kinetic energy of the stone.

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- (b) Determine the speed of the stone when it hits the slope.

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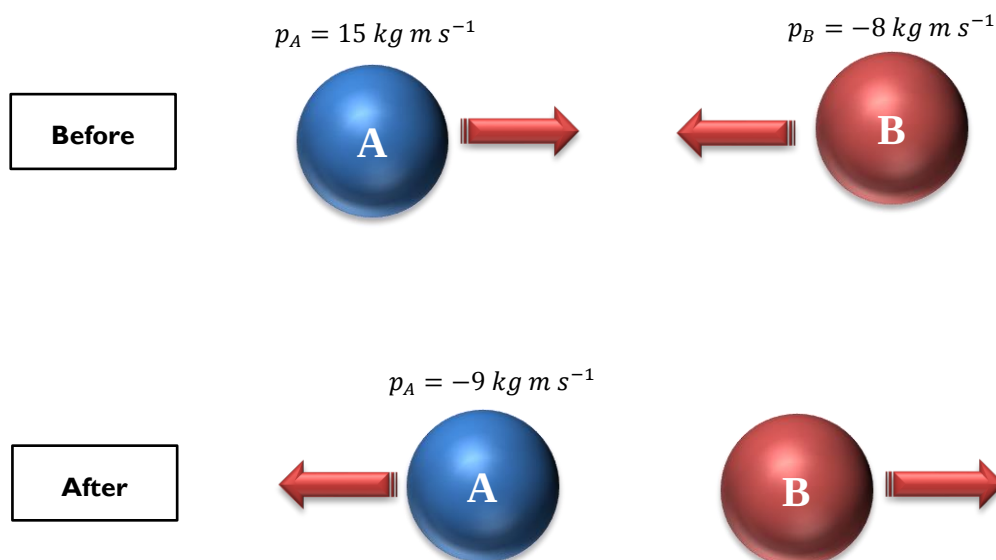
Class

 Name

Marks

Question 26 (6 marks)

Two balls, A and B (with momenta p_A and p_B), collide *elastically* as follows:



During the collision, ball A of mass 3 kg loses 24 joules of kinetic energy.

- (a) Calculate the magnitude of the final momentum p_B of ball B after the collision.

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- (b) Calculate the magnitude of the impulse that ball A applies to ball B during the collision.

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Question 26 continued on next page.

Question 26 continued.

Marks

(c) Determine the initial velocity of ball A.

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(d) Determine the mass of ball B.

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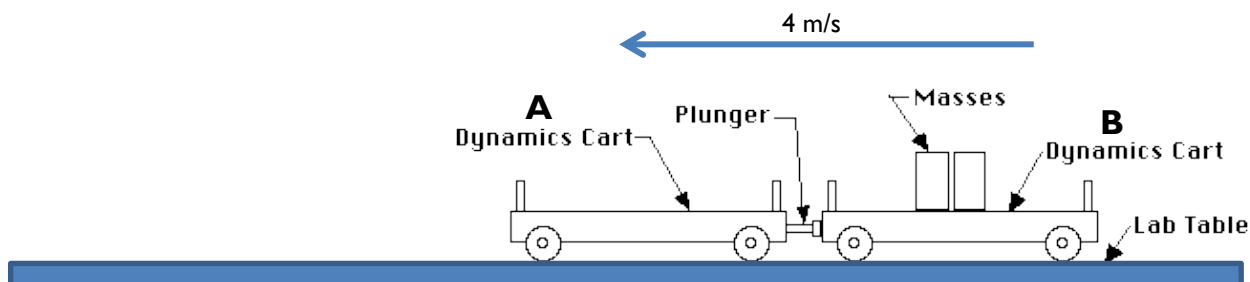
Class

Name

Marks

Question 27 (5 marks)

Two dynamics trolleys, A and B, are joined together and moving to the left at 4 m s^{-1} . The spring loaded plunger is set off pushing the trolleys apart. Trolley A increases in speed while trolley B becomes stationary.



The dynamics trolleys are 3 kg each and the two added masses to trolley B are 1 kg each.

- (a) Calculate the magnitude of the total momentum of trolley A and B before the plunger is set off.

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- (b) Determine the speed of trolley A after the plunger is set off.

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- (c) Show numerically whether the total kinetic energy increases, decreases or remains the same. Account for why this occurs.

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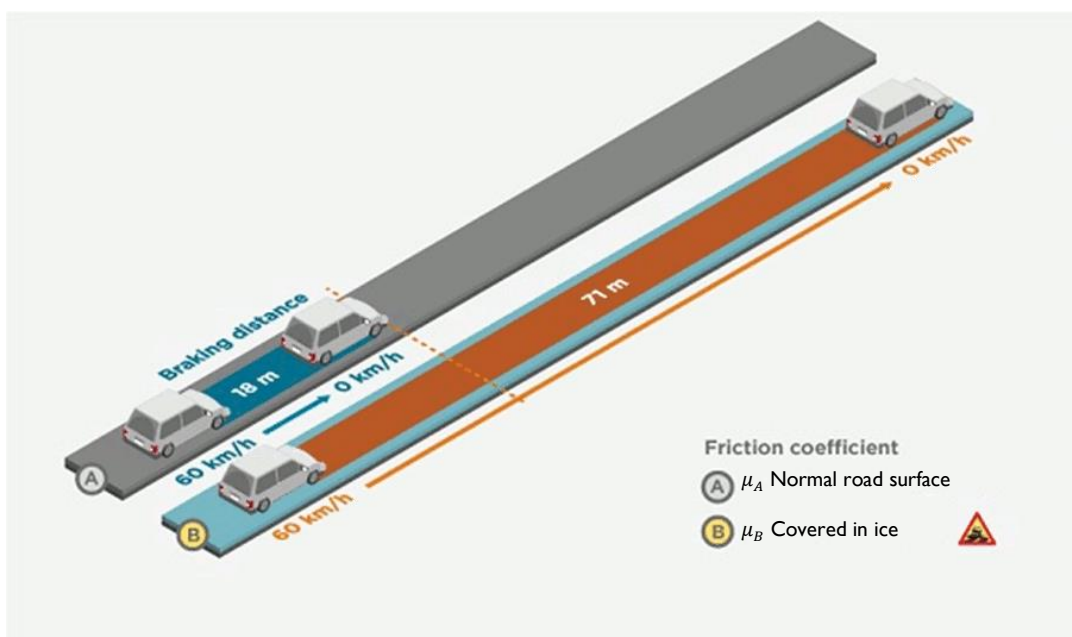
Question 28 (5 marks)**Marks**

The maximum braking force of a car is limited by the frictional force between the tyres and the road. The maximum frictional force, F , can be determined by the equation $F = \mu mg$ where μ is the coefficient of friction between the tyre and the road and mg is the weight of the car.

A car's braking distance is tested on two different road surfaces, A and B, both from an initial speed of 60 kmh^{-1} (16.67 ms^{-1}).

The braking distance on surface A was 18 m.

The braking distance on surface B was 71 m.



- (a) Determine the coefficient of friction for the normal road surface A.

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Question 28 continued on next page.

Class

Name

Question 28 continued.

Marks

- (b) Determine the maximum initial speed that should be travelled on road surface B to have the same stopping distance as surface A.

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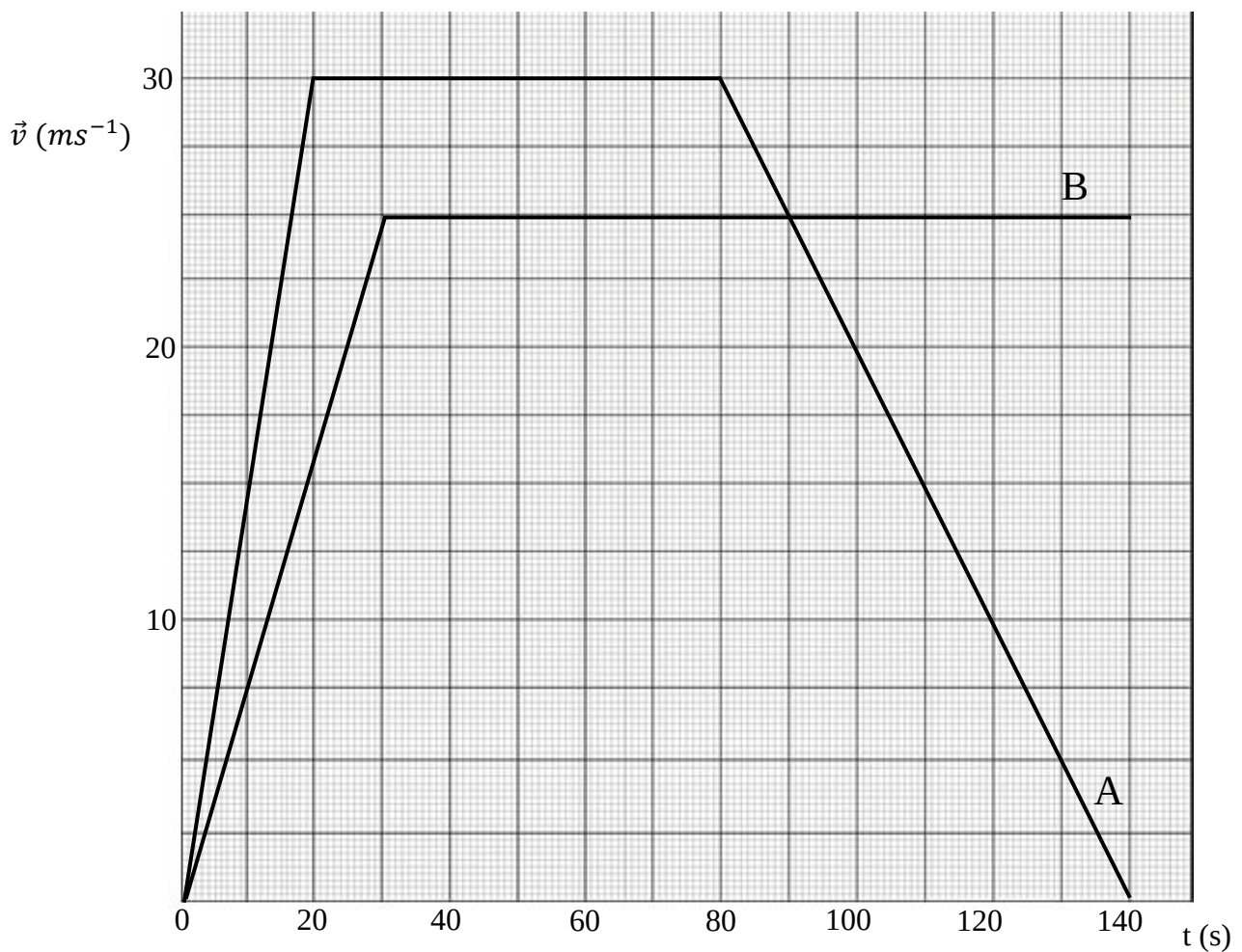
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ClassName

Marks

Question 29 (6 marks)

The velocity versus time graphs below show two underground trains, A and B, travelling from the same station on parallel, straight tracks. At time $t = 0$ s, both trains are alongside each other at rest.



Question 29 continued on next page.

Question 29 continued.

Marks

Using the graphs, or otherwise, determine:

- (a) the maximum speed of train B.

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- (b) the magnitude of the acceleration of train A between 80 and 140 s.

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- (c) the time at which train A and train B are alongside each other again.

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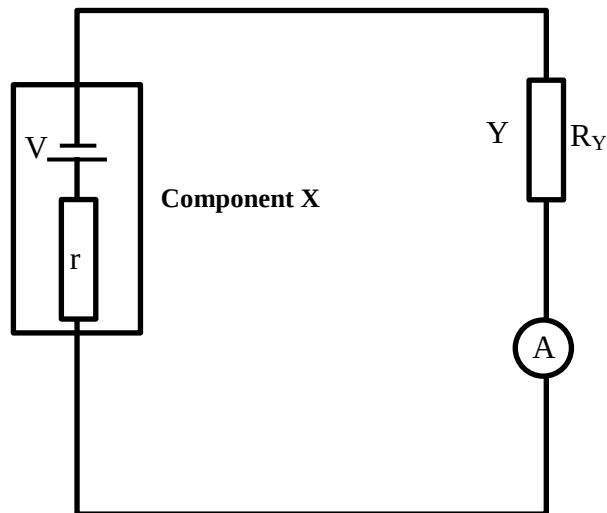
Class

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Name

Marks**Question 30** (4 marks)

A physics student sets up the following circuit in an attempt to determine the electrical characteristics of component X.



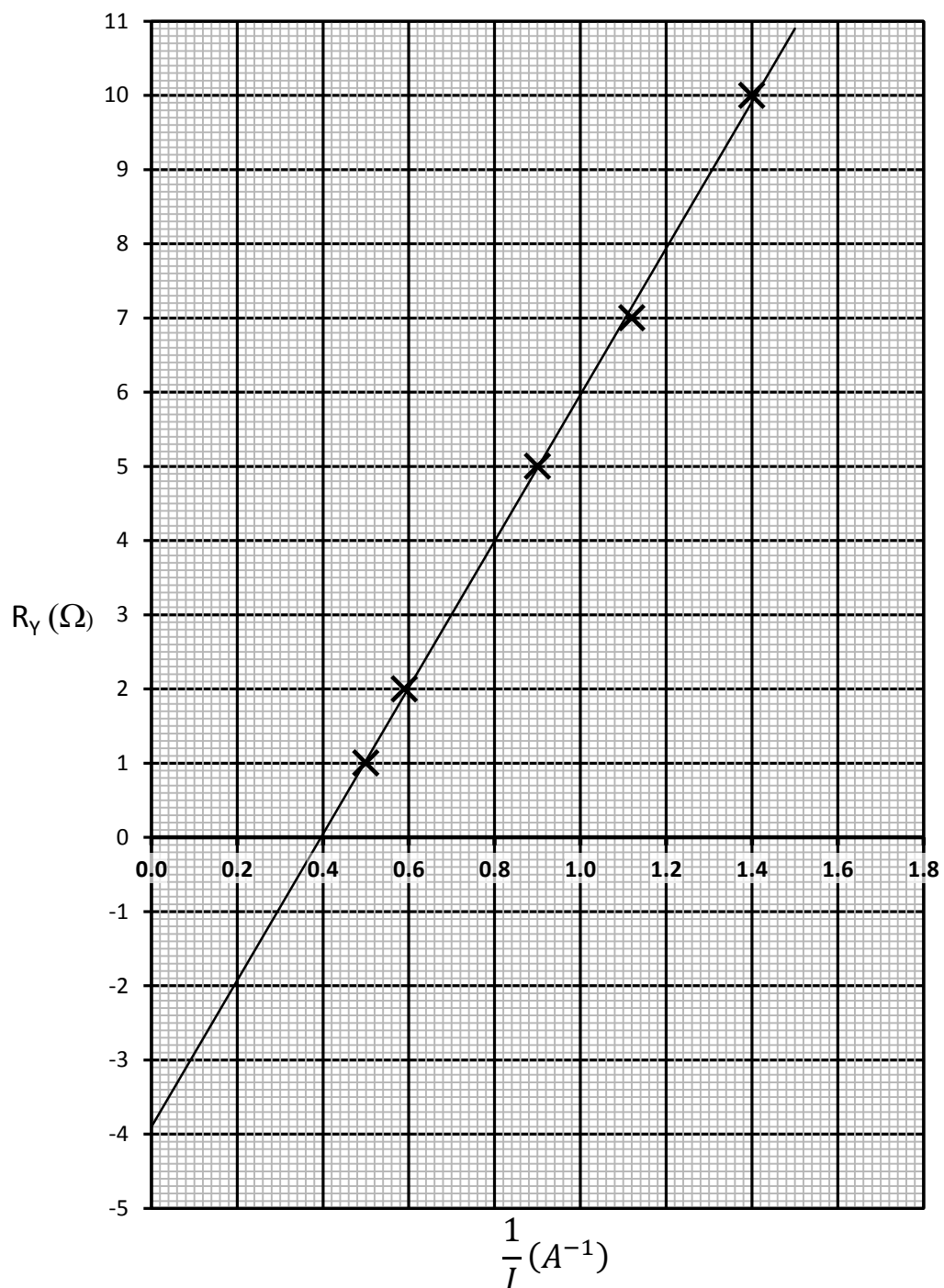
Component X consists of a battery of potential difference, V and a resistor, r , in series.

Various resistors, R_Y , are placed in the circuit at Y and the current flowing through the ammeter is recorded each time.

Question 30 continued on next page.

Question 30 continued.**Marks**

The following graph is a plot of R_Y (on the y-axis) versus $\frac{1}{I}$ using the experimental results obtained.



Question 30 continued on next page.

Class

Name

Question 30 continued.

Marks

- (a) Use the graph to find the value of the resistor, r , within component X.

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- (b) Use the graph to find the value of the potential difference, V , of the battery within component X.

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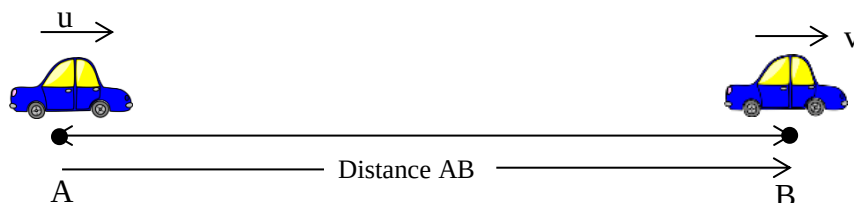
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Question 31 (10 marks)**Marks**

In an engine test, a car accelerates uniformly between two points (from the same starting velocity) on a straight road, and its final velocity is recorded.



The data obtained is included in the table on the following page.

Distance AB (m)	Final Velocity, v (ms^{-1})			Average v (ms^{-1})*	Average v^2 (m^2s^{-2})*
	1	2	3		
50	20.6	21.4	21.2	21.1	445
100	27.2	27.2	27.0	27.1	734
150	32.7	33.4	32.8	33.0	1090
200	37.3	37.5	37.1	37.3	1390
250	40.4	40.9	40.8		
300	44.2	44.6	44.8	44.5	1980

(* to 3 significant figures, with v^2 based on v to 3 significant figures)

- (a) Complete the table above by adding in the missing values.

Question 31 continued on next page.

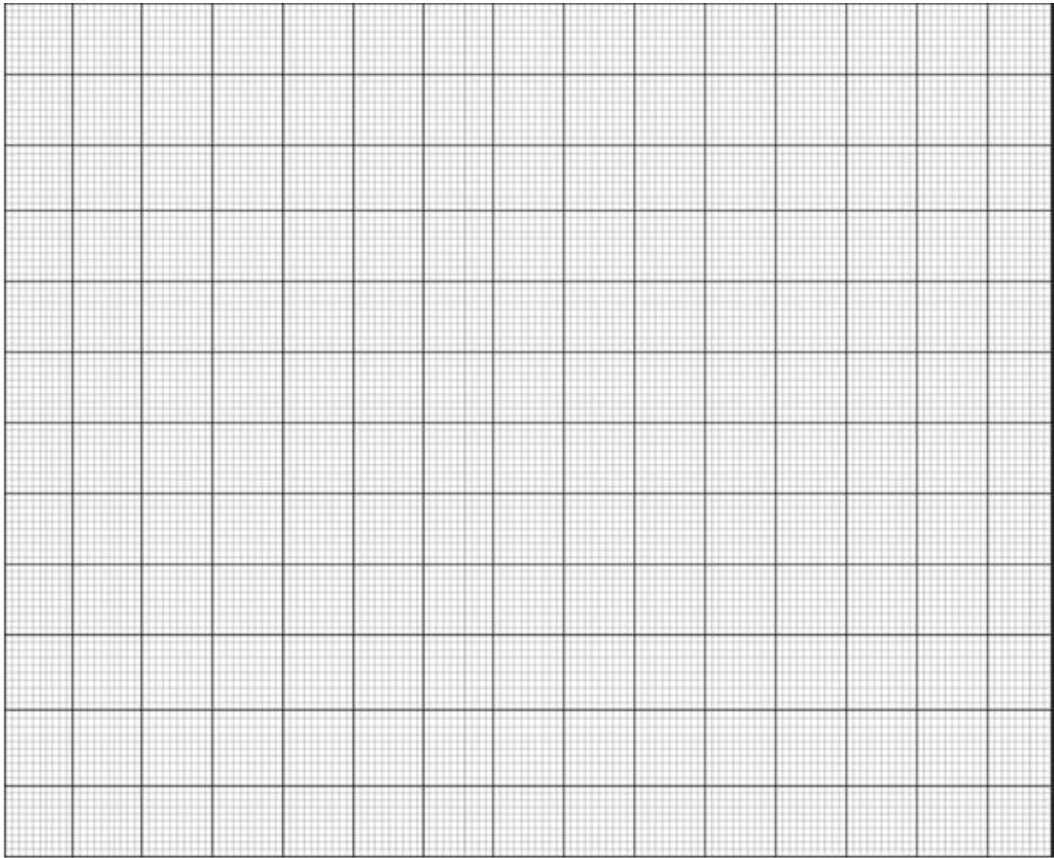
Class

Name

Question 31 continued.

Marks

(b) Plot a graph of velocity squared (on the y-axis) against distance.



4

Question 31 continued on next page.

Question 31 continued.

Marks

(c) Using your graph, or otherwise, determine:

(i) the initial velocity of the car.

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(ii) the acceleration of the car.

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Physics

Data Sheet

Charge on the electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Radius of Earth, R_E	$6.4 \times 10^6 \text{ m}$
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck's constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg's constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ 931.5 MeV/c^2
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET FORM V ONLY

$$v_{av} = \frac{\Delta r}{\Delta t}$$

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v-u}{t}$$

$$v = u + at$$

$$v^2 = u^2 + 2ar$$

$$r = ut + \frac{1}{2}at^2$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

$$W = Fr$$

$$p = mv$$

$$\Delta p = F_n t$$

$$F = mg$$

$$E = \frac{F}{q}$$

$$E = \frac{V}{d}$$

$$I = \frac{Q}{t}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v = f\lambda$$

$$f = \frac{1}{T}$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$n\lambda = d \sin \theta$$

$$n\lambda = \frac{dx}{L}$$

$$E_p = -\frac{Gm_1m_2}{r}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1m_2}{d^2}$$

$$E = mc^2$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{P}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100(m_B - m_A)/5$$

$$\tau = nBIa \cos \theta$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R_H \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$A_0 = \frac{V_{out}}{V_{in}}$$

$$c = f\lambda$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

$$Z = \rho v$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

$$\begin{array}{l} \text{Surface area of a} \\ \text{sphere of radius, } R \end{array} = 4\pi R^2$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY																2 He 4.003 Helium	
		Atomic Number Symbol Name																	
		Standard Atomic Weight																	
		Gold																	
		79 Au 197.0																	
		Cobalt																	
		27 Co 58.93																	
		Nickel																	
		28 Ni 58.69																	
		Copper																	
		29 Cu 63.55																	
		Zinc																	
		30 Zn 65.38																	
		Gallium																	
		31 Ga 69.72																	
		Germanium																	
		32 Ge 72.64																	
		Arsenic																	
		33 As 74.92																	
		Selenium																	
		34 Se 78.96																	
		Bromine																	
		35 Br 79.90																	
		Krypton																	
		36 Kr 83.80																	
		Xenon																	
		54 Xe 131.3																	
		Iodine																	
		53 I 126.9																	
		Tellurium																	
		52 Te 127.6																	
		Polonium																	
		84 Po 209.0																	
		Astatine																	
		85 At 209.0																	
		Radon																	
		86 Rn 222.0																	
		Francium																	
		87 Fr 223.0																	
		Radium																	
		88 Ra 226.0																	
		Actinoids																	
		89-103																	
		Lanthanoids																	
		57-71																	
		Hf																	
		72 Hf 178.5																	
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		78 Pt 195.1																	
		Au																	
		79 Au 197.0																	
		Ag																	
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		49 In 114.8																	
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		50 Sn 118.7																	
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		51 Sb 121.8																	
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		Xe																	
		54 Xe 131.3																	
		Rn																	
		86 Rn 222.0																	
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		87 Fr 223.0																	
		Radium																	
		88 Ra 226.0																	
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		61 Pm 144.9																	
		Sm																	
		62 Sm 150.4																	
		Eu																	
		63 Eu 152.0																	
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		71 Lu 175.0																	
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KEY

Atomic Number	79
Symbol	Au
Standard Atomic Weight	197.0
Name	Gold

Elements with atomic numbers 112 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.



Class

Name

PHYSICS
Form V
May 2016
Monday 23rd MAY 1.00 PM

General Instructions

- Write your class and Candidate Number in the space provided.
- Attempt all questions
- Use a blue or black pen
- Select the alternative A, B, C, or D that best answers the question.
- Fill in the response oval completely.

Multiple Choice
ANSWER SHEET

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | ☐ A | ☐ B | ☐ C | ☐ D |
| 2. | ☐ A | ☐ B | ☐ C | ☐ D |
| 3. | ☐ A | ☐ B | ☐ C | ☐ D |
| 4. | ☐ A | ☐ B | ☐ C | ☐ D |
| 5. | ☐ A | ☐ B | ☐ C | ☐ D |
| 6. | ☐ A | ☐ B | ☐ C | ☐ D |
| 7. | ☐ A | ☐ B | ☐ C | ☐ D |
| 8. | ☐ A | ☐ B | ☐ C | ☐ D |
| 9. | ☐ A | ☐ B | ☐ C | ☐ D |
| 10. | ☐ A | ☐ B | ☐ C | ☐ D |
| 11. | ☐ A | ☐ B | ☐ C | ☐ D |
| 12. | ☐ A | ☐ B | ☐ C | ☐ D |
| 13. | ☐ A | ☐ B | ☐ C | ☐ D |

Mark

MTK.
Class

CRIB.

Name

Part B**Total marks (86)****Attempt ALL Questions****Allow about 1 hour and 45 minutes for this Part**

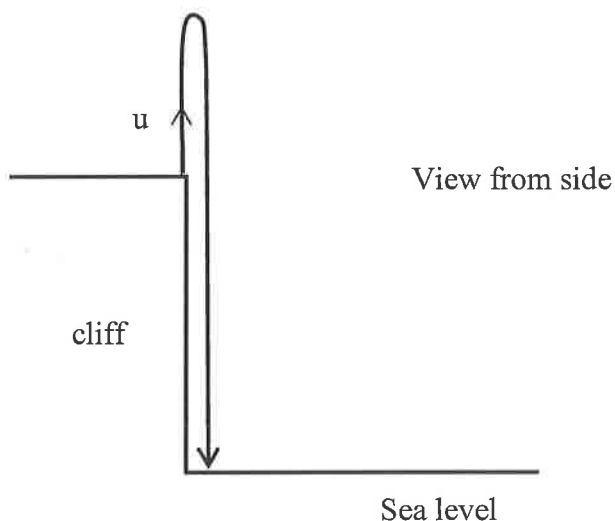
Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 14 (5 marks)**Marks**

A ball was thrown vertically upwards at a speed of 10.0 m s^{-1} from the top of a cliff. The time of flight for the ball to hit the sea at the bottom of the cliff was 6.92 s.

Assume no air resistance acts on the ball.



- (a) Calculate the speed with which the ball hits the sea at the bottom of the cliff.

$$\begin{aligned}
 v &= u + at \\
 &= 10 + -9.8 \times 6.92 \\
 &= 57.8 \text{ ms}^{-1} \downarrow
 \end{aligned}$$

2

Question 14 continued on next page.

Question 14 continued.

Marks

- (b) Calculate the height of the cliff top above sea level.

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ &= 10 \times 6.92 + \frac{1}{2} \times -9.8 \times 6.92^2 \\ &= 165.4 \text{ m} \end{aligned}$$

2

- (c) What is the acceleration of the ball at the highest point?

$$-9.8 \text{ ms}^{-2} \downarrow$$

1

Class

 Name

Marks

Question 15 (6 marks)

A police motor bike is stationary by the side of a road. An L-plate driver, exceeding the speed limit, passes the police motor bike at a constant speed of 108 km h^{-1} . The police motor bike sets off to catch the L-plater just as he passes. The police motor bike accelerates uniformly at 6.5 m s^{-2} for 6.0 s and then continues at a constant speed.

- (a) Convert the L-plate driver's speed to
- ms^{-1}
- .

$$108 / 3.6 = 30 \text{ ms}^{-1}$$

1

- (b) Calculate the speed of the police motor bike at
- 6 s
- .

$$v = u + at = 0 + 6.5 \times 6 = 39 \text{ ms}^{-1}$$

1

- (c) Calculate the distance travelled by police motor bike in the first
- 6 s
- .

$$r = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 6.5 \times 6^2 = 117 \text{ m}$$

1

- (d) Calculate the total time for the police motor bike to draw level with the L-plate driver.

Car in 6 s $6 \times 30 = 180 \text{ m} \therefore \text{must be after } \frac{1}{2} 6 \text{ s}.$

$$(1) \quad 30t = 117 + 39 \times (t - 6)$$

$$(1) \quad t = 13 \text{ s}$$

(OR) $(1) \quad 180 - 117 = 63 \quad 39 - 30 = 9 \quad 63/9 = 7 \text{ s}.$

$$(1) \quad \text{total} = 6 + 7 = 13 \text{ s}$$

- (e) Calculate how far the L-plate driver has travelled before the police motor bike catches up.

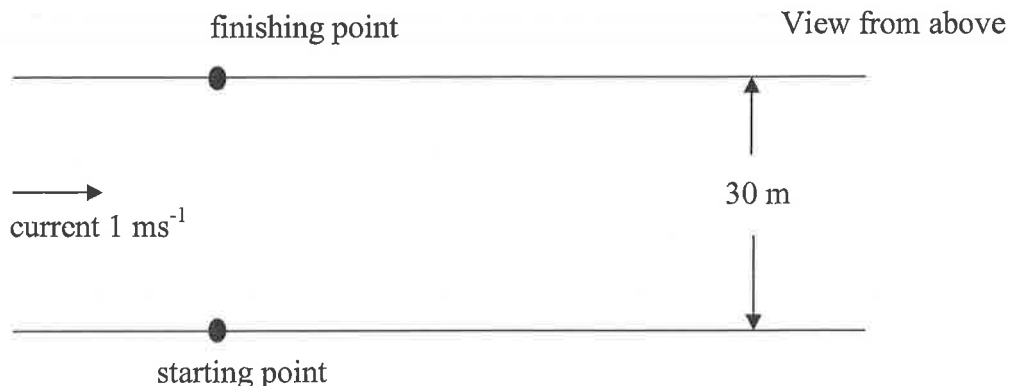
$$r = ut = 30 \times 13 = 390 \text{ m}$$

↑
or Ans(d)

1

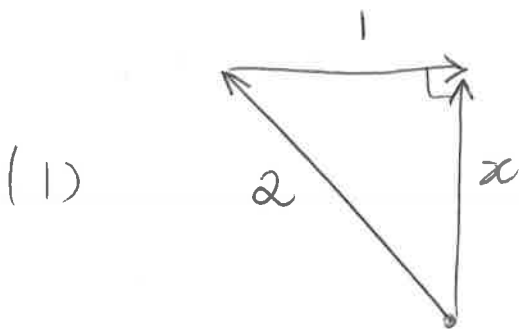
Question 16 (3 marks)**Marks**

A swimmer competes in a swimming race across a 30 m wide river which has a constant downstream current of 1.0 ms^{-1} . He swims slightly against the current so that he ends up exactly opposite the starting point as shown in the following diagram.



The swimmer can swim at 2 ms^{-1} in still water.

By drawing velocity vectors in the space provided below, determine the time it takes the swimmer to swim from the starting point to the finishing point.



$$(1) \quad 1^2 + x^2 = 2^2$$

$$x = \sqrt{3} \text{ ms}^{-1}$$

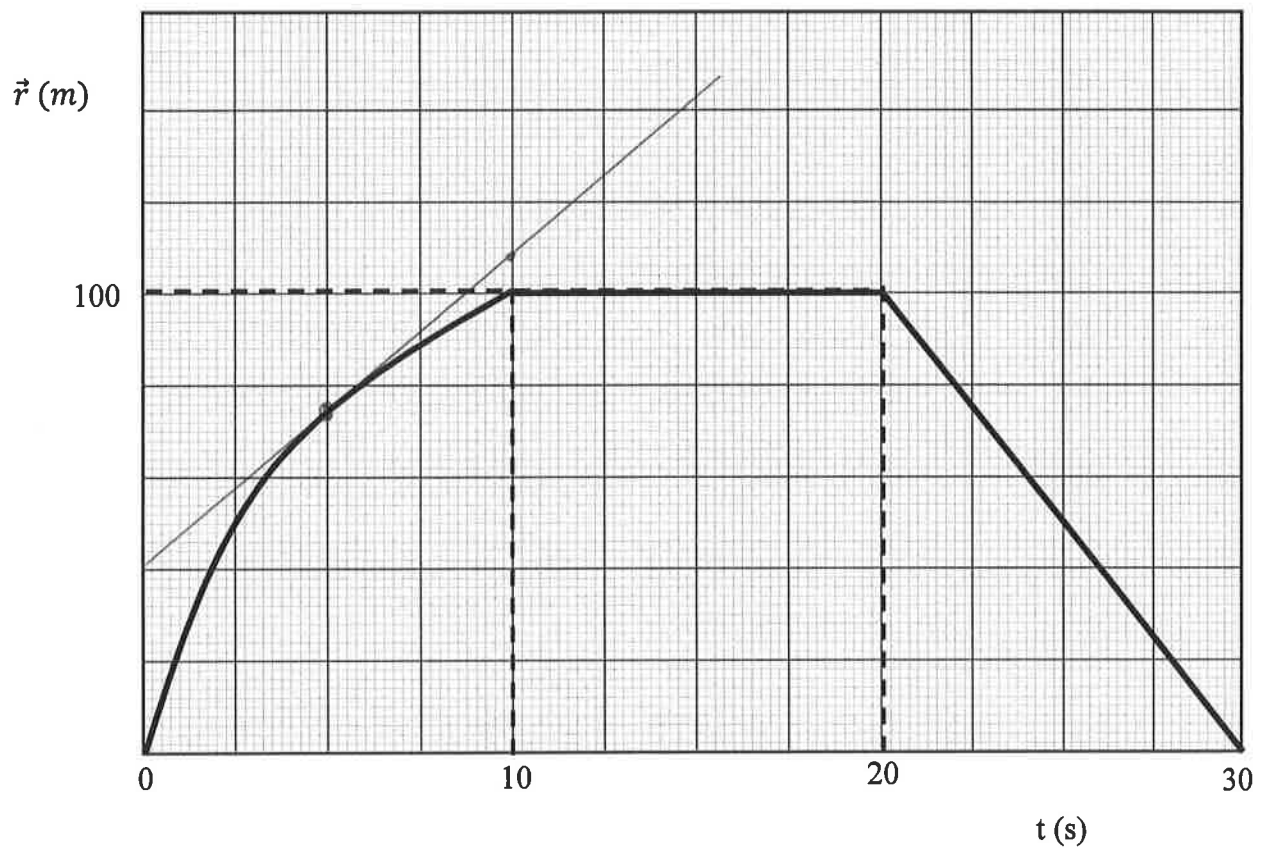
$$(1) \quad t = \frac{d}{v} = \frac{30}{\sqrt{3}} = 17.3 \text{ s}$$

3

Class

 Name
Question 17 (5 marks)**Marks**

The diagram below shows the displacement versus time graph for a car travelling along a straight road.



From the graph, determine:

- (a) the average speed of the car between 0 and 30 s.

$$\text{speed} = d/t = (100 + 100)/30$$

$$= \underline{6.7 \text{ ms}^{-1}}$$

1

Question 17 continued on next page.

Question 17 continued.

Marks

- (b) the average velocity of the car between 20 and 30 s.

$$\begin{aligned}
 v_{AV} &= \frac{r}{t} \\
 &= \frac{0 - 100}{10} \\
 &= -10 \text{ ms}^{-1}
 \end{aligned}$$

(1) (1)

OR backwards

- (c) the instantaneous velocity of the car at 5 s.

(1) - recognising tangent required.

(1) - calculating gradient of tangent.

$$v = \frac{106 - 40}{10} = 6.6 \text{ ms}^{-1}$$

* paid a wide range of answers as long as it matched the tangent drawn.

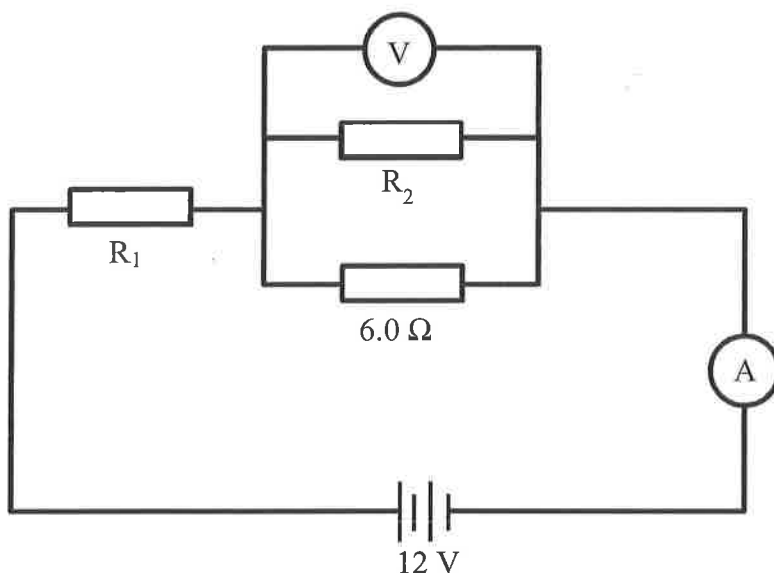
(1) only for $v_{AV} = \frac{74}{5} = 14.8 \text{ ms}^{-1}$.

Class

Name

Question 18 (4 marks)**Marks**

This question refers to the electric circuit shown below.



The voltmeter has a reading of 4.0 V and the ammeter has a reading of 3.0 A.

- (a) Calculate the voltage across the unknown resistor, R_1 .

$$\begin{aligned} V_T &= V_{R_1} + V_{R_2} \\ 12 &= V_{R_1} + 4 \\ V_{R_1} &= 8V \end{aligned} \quad \left. \begin{array}{l} \text{✓ ①mk for} \\ \text{correct method.} \end{array} \right\} 1$$

- (b) Calculate the value of the unknown resistor, R_1 .

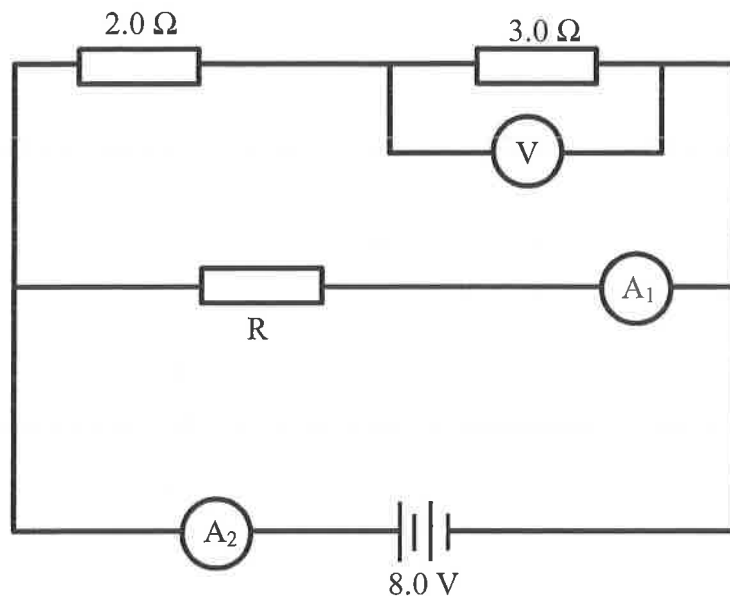
$$R_1 = \frac{V_{R_1}}{I} = \frac{\text{Ans(a)}}{3} = \frac{8}{3} = 2.67 \Omega \quad \left. \begin{array}{l} \text{✓ ①mk} \end{array} \right\} 1$$

- (c) Calculate the value of the unknown resistor, R_2 .

$$\begin{aligned} \text{Step 1: } V &= IR \\ 12 &= 3R \\ R_1 &= 4 \Omega \\ \text{Step 2: } R_T &= R_1 + R_2 \\ 4 &= \text{Ans(a)} + R_2 \\ 4 &= 2.67 + R_2 \\ R_2 &= 1.33 \Omega \\ \text{Step 3: } \frac{1}{R_2} &= \frac{1}{R_1} + \frac{1}{R_2} \\ \frac{1}{1.33} &= \frac{1}{6} + \frac{1}{R_2} \\ R_2 &= 1.71 \Omega \quad \left. \begin{array}{l} \text{Method 2 based} \\ \text{on finding} \\ \text{total } R, \\ \text{then } R_2 \\ \text{then} \\ \frac{1}{R_2} = \frac{1}{R_1} + \frac{1}{R_2} \\ \text{but} \\ \text{answer} \\ \text{incorrect} \\ \text{✓ x ①mk.} \end{array} \right\} 2 \text{ mks.} \end{aligned}$$

Question 19 (4 marks)**Marks**

This question refers to the electric circuit shown below.

The reading on the ammeter, A_1 is 0.80 A.

- (a) Calculate the value of the unknown resistor,
- R
- .

$$R = V/I = 8/0.8 \quad \checkmark \text{ ①mk}$$

$$= 10 \Omega$$

substitution
into correct
formula

1

- (b) Determine the reading on the voltmeter in the circuit above.

$$I_{3\Omega} = \frac{V}{R} = 8/5 = 1.6 \text{ A} \quad \checkmark \text{ ①mk}$$

$$V_{3\Omega} = 3 \times 1.6$$

$$= 4.8 \text{ V} \quad \checkmark \text{ ①mk}$$

i.e. $V = 3 \times I$
↑
correct
or
incorrect
current.

2

- (c) Calculate the reading on the ammeter
- A_2
- .

$$I_2 = I_1 + I_{3\Omega}$$

$$= 0.8 + \text{Answer for } I \text{ used in (b)}$$

$$= 0.8 + 1.6 = 2.4 \text{ A}$$

1

OR/

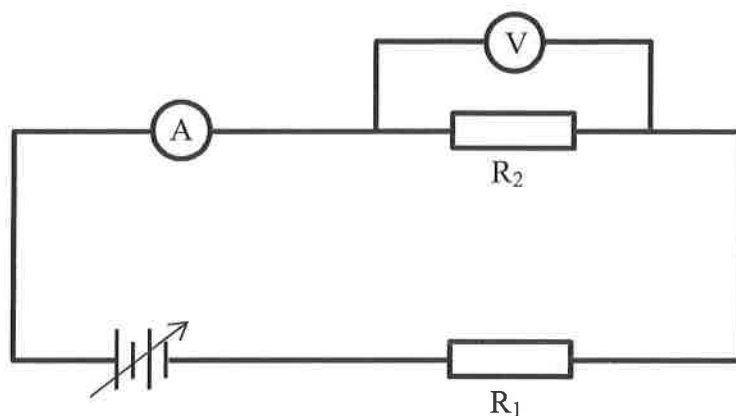
$$\frac{V_T}{R_T} = I_T = A_2$$

← use values from (a) and question to calculate R_T correctly

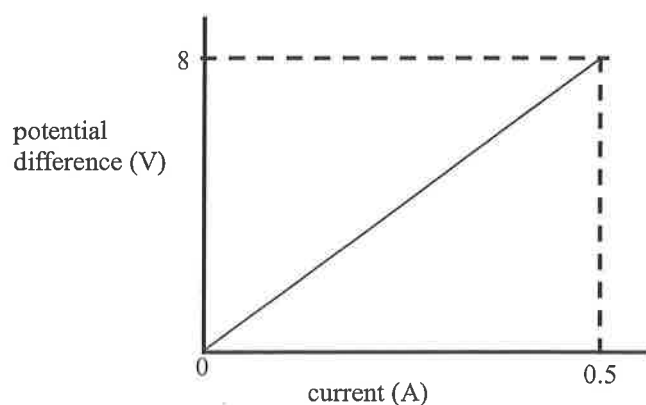
Class

 Name
Question 20 (3 marks)**Marks**

The following circuit was set up with two resistors, R_1 and R_2 , and a battery.



The following graph shows how the potential difference across resistor, R_2 , varies as the current through resistor, R_2 , is changed.



- (a) Calculate the value of the resistance of resistor, R_2 .

$$R = \frac{V}{I} = \frac{8}{0.5} = 16 \Omega$$

1
 ✓ 1mk for substitution into correct formula.

Question 20 continued on next page

Question 20 continued.

Marks

- (b) When the potential difference of the battery is 8.0 V, it supplies a current of 0.40 A to the circuit.

Calculate the value of the unknown resistor, R_1 .

$$R_T = 8/0.4 = 20 \Omega \quad \checkmark \text{ 1mk if calculated total resistance only} \quad 2$$

$$R_T = R_1 + R_2$$

$$20 = \text{Ans(a)} + R_2 \quad \checkmark \text{ 1mk}$$

$$R_2 = 4 \Omega \quad \checkmark R_2 = 20 - \text{Ans(a)}$$

OR/ $8 = (16 + R_1) \times 0.40 \quad \checkmark \text{ 2mks}$

OR/ calculate V across R_1 $\checkmark \text{ 1mk}$
 $V = 0.4 \times \text{Ans(a)} \quad \checkmark \text{ 1mk}$

Question 21 (2 marks) $V_{R_1} = 8 - 6.4$ then $V_{R_1} = R \times 0.4 \quad \checkmark \text{ 1mk}$

A 100 kg man stands on a set of bathroom scales in a lift and notices that the scales read 153 kg.

Determine the acceleration of the lift.

$$\Sigma F = ma \quad \checkmark \text{ 1mk} \quad 2$$

$$153 \times 9.8 - 100 \times 9.8 = 100a$$

$$a = 5.2 \text{ ms}^{-2} \quad \checkmark \text{ 1mk}$$

OR/ $153 \times 9.8 = 100a$ weight not included $\checkmark \text{ 1mk}$
 $a = 14.99 \text{ ms}^{-2}$

OR/ $153 - 100 \times 9.8 = 100a$ R not converted to newtons $\checkmark \text{ 1mk}$

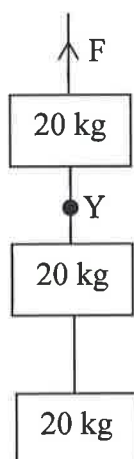
OR/ $153 \times 9.8 - 100 \times 9.8 = 153a \quad \checkmark \text{ 1mk}$

OR/ $153 \times 9.8 = 100a \quad \checkmark \text{ 1mk}$

SRW	Class
Name	

Question 22 (4 marks)**Marks**

Three 20 kg masses are pulled **upwards** with a constant force F such that the tension force in the rope at point Y is 500 N.



Many boys
left out gravity
Maximum marks = $\frac{2}{4}$

- (a) Determine the magnitude of the acceleration of the masses.

$$500 - 40g = 40a \quad (1)$$

2

$$a = 2.7 \text{ m/s}^2 \quad (1)$$

- (b) Determine the magnitude of the constant force, F .

$$F - 60g = 60a \quad (1)$$

2

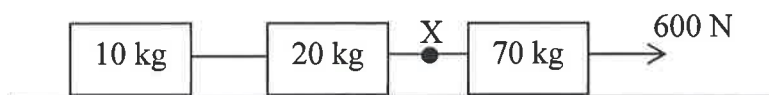
$$F = 60(a + g)$$

$$= 750 \text{ N} \quad (1)$$

Many boys got 750 N even though they ignored gravity. They only got a maximum of $\frac{2}{4}$ marks

Question 23 (4 marks)**Marks**

Three masses 10 kg, 20 kg and 70 kg are connected by massless and unstretchable strings and pulled along a frictionless, horizontal surface by a constant horizontal force of 600 N as shown below.



- (a) Calculate the magnitude of the acceleration of the masses.

$$a = \frac{600}{100} = 6 \text{ m/s}^2 \quad (1)$$

1

- (b) Calculate the tension force in the string at X.

$$T = 30a = 180 \text{ N} \quad (1)$$

1

- (c) The same arrangement of three masses is now placed on a rough surface. If the frictional force which now acts on each mass is equal to 0.2 times the weight of each mass, determine the magnitude of the new acceleration of the system when the constant horizontal force of 600 N is applied.

$$600 - 0.2 \times 100 \times g = 100a \quad (1)$$

2

$$a = 4.04 \text{ m/s}^2 \quad (1)$$

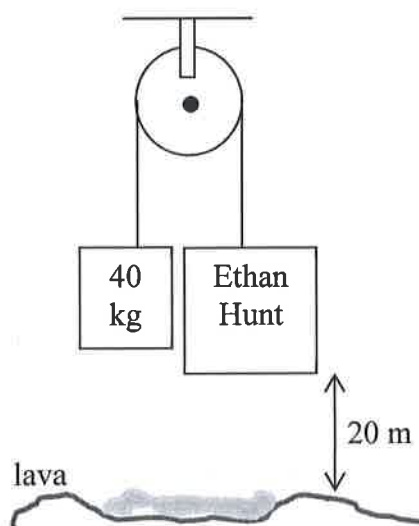
Manny boys
forgot "g"
I took off 1 mark

Class

Name

Question 24 (6 marks)**Marks**

In “*Mission Impossible Twelve*”, Ethan Hunt of mass 80 kg finds himself attached to a 40 kg mass via a pulley above a pool of lava as shown below.



- (a) Assuming the system is released from rest, what time would elapse before Ethan reaches the surface of the lava?

$$a = \frac{40g}{120} = 3.26 \text{ m/s}^2 \text{ down } \textcircled{1}$$

2

$$r = \frac{1}{2}at^2 \quad \therefore t = \sqrt{\frac{2 \times 20}{3.26}} = 3.5 \text{ s } \textcircled{1}$$

(allowed carry through error)

- (b) Ethan starts to climb up the rope with an acceleration of 1.0 m s^{-2} **relative to the lava**.

- (i) Determine the tension in the rope while Ethan is climbing.

$$T - 80g = 80a \textcircled{1}$$

2

$$T = 80(g + 1) = 864 \text{ N } \textcircled{1}$$

Question 24 continued on next page.

Question 24 continued.

Marks

- (ii) Determine the magnitude of the acceleration of the 40 kg mass while Ethan climbs the rope.

$$T - 40g = 40a \quad (1)$$

2

$$\therefore a = \frac{864 - 40g}{40} = 11.8 \text{ m/s}^2 \quad (1)$$

← from part (i)

$$a = \frac{T - 40g}{40}$$

Carry
through
error
allowed.

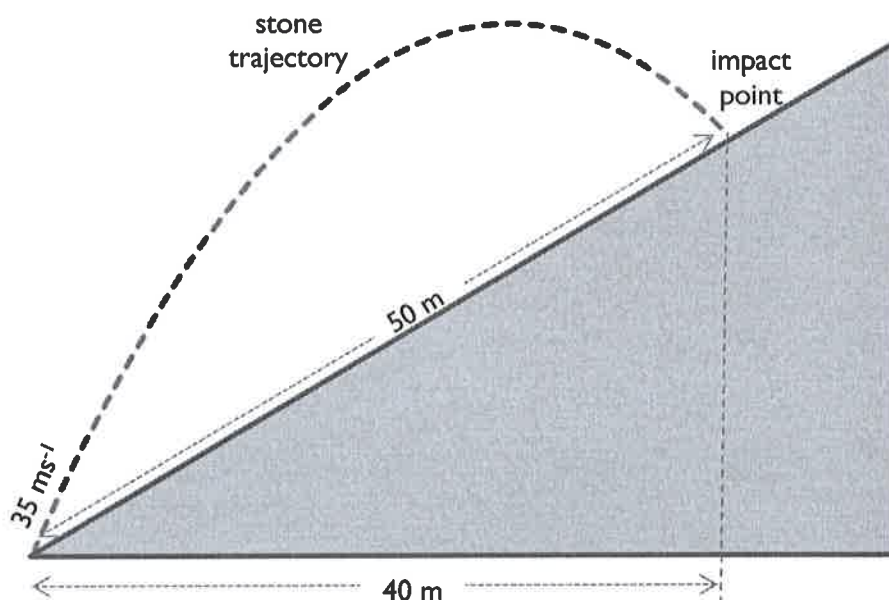
Class

Name

Marks

Question 25 (4 marks)

A stone of mass 0.25 kg is thrown into the air with an initial speed of 35 m s^{-1} and it lands 50 m up a slope as shown in the following diagram.



- (a) Determine the initial kinetic energy of the stone.

$$\frac{1}{2}mv^2 = \frac{1}{2} \times 0.25 \times 35^2 = 153.125 \text{ J.} \quad (1)$$

1

- (b) Determine the speed of the stone when it hits the slope.

$$\text{Final Energy} = \text{GPE} + \text{KE}$$

$$153.125 = 73.5 + \text{KE} \quad (1)$$

$$\therefore \text{KE} = 79.625$$

$$\therefore \frac{1}{2}mv^2 = 79.625 \quad (1)$$

$$\therefore v = \sqrt{637} = 25.2 \text{ m/s} \quad (1)$$

3

Many boys used a weird method where they assumed the initial vertical velocity was 35 m/s . I only gave (2) marks for this wrong physics

CRIB

Class

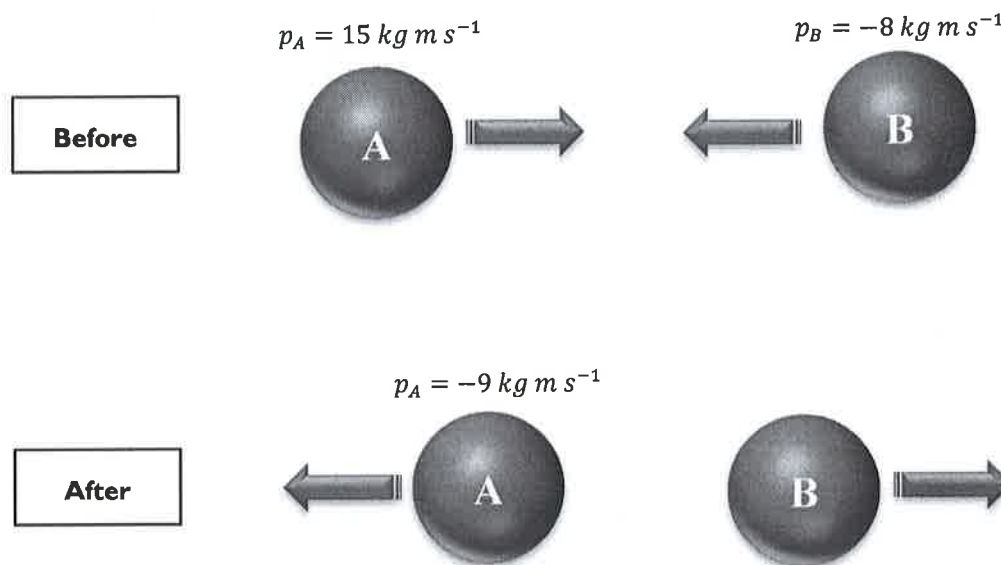
PCK

Name

Marks

Question 26 (6 marks)

Two balls, A and B (with momenta p_A and p_B), collide *elastically* as follows:



During the collision, ball A of mass 3 kg loses 24 joules of kinetic energy.

- (a) Calculate the magnitude of the final momentum p_B of ball B after the collision.

$$\sum p_i = 15 - 8 = 7 \text{ kg m/s}$$

$$\sum p_f = -9 + p_B = 7 \quad p_B = 16 \text{ kg m/s}$$

1

- (b) Calculate the magnitude of the impulse that ball A applies to ball B during the collision.

$$I = \Delta p_B = p_{B(\text{final})} - p_{B(\text{initial})} = 16 - (-8)$$

Impulse is the change in momentum. $= 24 \text{ Ns}$

The Impulse on ball B is the change of momentum of ball B.

1

Question 26 continued on next page.

Question 26 continued.

Marks

- (c) Determine the initial velocity of ball A.

given $m_A = 3 \text{ kg}$ $P_A(\text{initial}) = m_A u_A = 15$, $u_A = \frac{15}{3}$

1

$u_A = 5 \text{ m/s}$
(positive, or to the right direction).

- (d) Determine the mass of ball B.

For an elastic collision $\sum KE_i = \sum KE_f$.

Similar to c) final velocity of ball A can be found $v_A = \frac{P_A(\text{final})}{3} = -3 \text{ m/s}$.

Since m_B is unknown, but momenta are $u_B = \frac{-8}{m_B}$, $v_B = \frac{16}{m_B}$.

Kinetic Energy can also be written $KE_B = \frac{1}{2} \left(\frac{P_B}{m_B} \right)^2 m_B = \frac{1}{2m_B} P_B^2$

3

Method 1.
use $\sum KE$
with v_A

Mo. (1/2) (A) Before (B) (A) After (B)
 $3(5)^2 + \frac{1}{m_B}(-8)^2 = 3(-3)^2 + \frac{1}{m_B}(16)^2$
 $75 + \frac{64}{m_B} = 27 + \frac{256}{m_B}$
 $\frac{1}{m_B}(256 - 64) = 75 - 27$
 $m_B = \frac{192}{48}$
 $m_B = 4 \text{ kg}$

Method 3 Relative velocities.
In elastic collision

$v_B - v_A = u_A - u_B$
 $\frac{16}{m_B} - (-3) = 5 - \left(-\frac{8}{m_B}\right)$
 $\frac{8}{m_B} = 2$
 $m_B = 4 \text{ kg}$

Method 2
use $\Delta KE = 24 \text{ J}$

$\frac{1}{2} 3(5)^2 + \frac{1}{2m_B}(-8)^2 = \left[\frac{1}{2} 3(5)^2 - 25 \right] + \frac{1}{2m_B}(16)^2$
 $37.5 + \frac{32}{m_B} = 13.5 + \frac{128}{m_B}$
 $\frac{1}{m_B}(128 - 32) = 24$
 $m_B = \frac{96}{24} = 4 \text{ kg}$

Method 4. Centre of mass

$V_{cm} = \frac{\sum p}{\sum m} = \frac{7}{3+m_B}$
 $v_A = 2(V_{cm}) - u_A$

$-3 = \frac{14}{3+m_B} - 5$
 $\frac{14}{3+m_B} = 2$, $m_B + 3 = 14\left(\frac{1}{2}\right) = 7$
 $m_B = 7 - 3 = 4 \text{ kg}$

ALL SOLUTIONS		A	B
Before	Momentum	15	-8
	velocity	5	-2
	KE	37.5	8
After	Momentum	-9	16
	velocity	-3	4
	KE	13.5	32
	ΔKE	-24	+24

Only 15%
of form
could
solve

Marking Uses relations
between mass, momentum
and velocity.

① Valid Method $\sum KE_i = \sum KE_f$
 ① Correct answer solvable equation

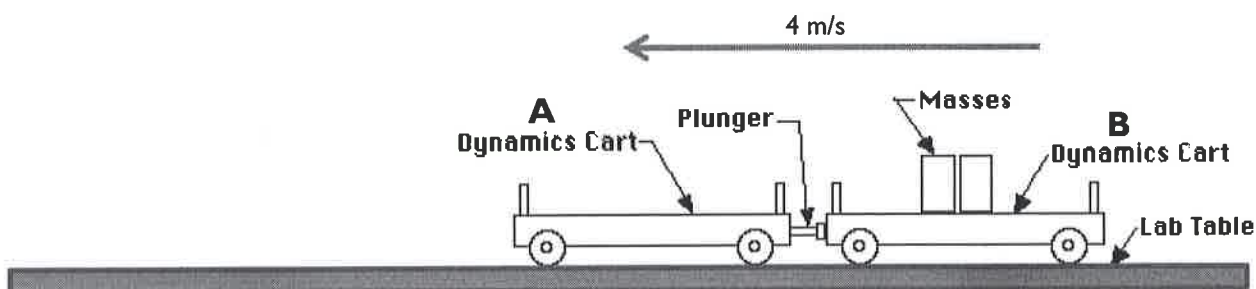
Class

Name

Marks

Question 27 (5 marks)

Two dynamics trolleys, A and B, are joined together and moving to the left at 4 m s^{-1} . The spring loaded plunger is set off pushing the trolleys apart. Trolley A increases in speed while trolley B becomes stationary.



The dynamics trolleys are 3 kg each and the two added masses to trolley B are 1 kg each.

- (a) Calculate the magnitude of the total momentum of trolley A and B before the plunger is set off.

$$\Sigma p_i = m_A u_A + m_B u_B = 4(m_A + m_B) = -4(3 + 5) = -32 \text{ kg m/s}$$

32 kg m/s

1

- (b) Determine the speed of trolley A after the plunger is set off.

$$v_B = 0, \Sigma p_f = m_A v_A = 32$$

$$v_A = \frac{32}{3} = 10.7 \text{ m/s}$$

(3 sig fig) (11 m/s to 2 sig fig)

1

- (c) Show numerically whether the total kinetic energy increases, decreases or remains the same. Account for why this occurs.

$$\Sigma KE_i = \frac{1}{2} 8(4)^2 = 64 \text{ J} \quad \Sigma KE_f = \frac{1}{2} (3)(10.67)^2 = 171 \text{ J}$$

3

① mark for KE_i , KE_f both correct
 total KE increases. ① must state explicitly
 total KE increases because there is stored potential energy in the spring that does work when released.
 ① Identifies that additional energy is coming from the spring

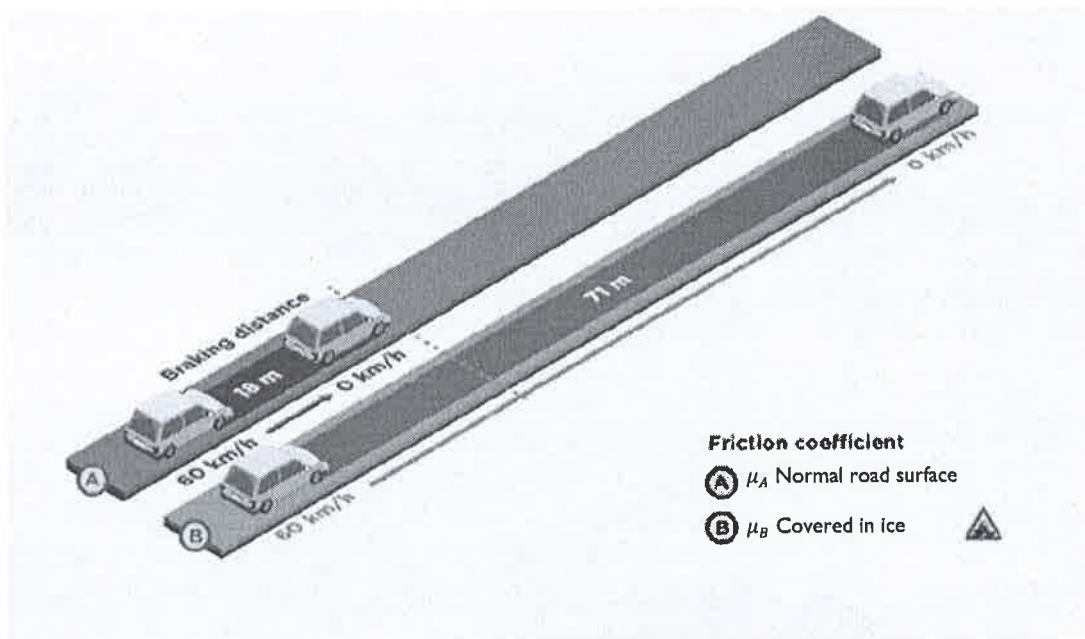
Question 28 (5 marks)**Marks**

The maximum braking force of a car is limited by the frictional force between the tyres and the road. The maximum frictional force, F , can be determined by the equation $F = \mu mg$ where μ is the coefficient of friction between the tyre and the road and mg is the weight of the car.

A car's braking distance is tested on two different road surfaces, A and B, both from an initial speed of 60 kmh^{-1} (16.67 ms^{-1}).

The braking distance on surface A was 18 m.

The braking distance on surface B was 71 m.



- (a) Determine the coefficient of friction for the normal road surface A.

Method 1.

Energy

Method 2.

Force

$$F d_A = (\mu_A mg) d_A = \frac{1}{2} m u^2, \mu_A = \frac{u^2}{2g d_A} = \frac{(16.67)^2}{2 \times 9.81 \times 18}$$

① equates energy

$$\text{Find } a = \frac{v^2 - u^2}{2r} = \frac{-u^2}{2r} = \frac{-u^2}{2 \times 18} = -7.72 \text{ m/s}^2, F = ma = \mu_A mg, \mu_A = \frac{a}{g} = \frac{-7.72}{-9.81}$$

OR [① acceleration, or ① equates forces]

coefficient of friction.

Question 28 continued on next page.

$\mu_A = 0.79$ or 0.8
(unitless) and ① mark - correct coefficient

Class

Name

Question 28 continued.

Marks

- (b) Determine the maximum initial speed that should be travelled on road surface B to have the same stopping distance as surface A.

Method 1.

Find μ_B
on ice.

$$\mu_B = \frac{(16.67)^2}{2 \times 9.8 \times 71} = 0.2 \quad \text{① Finds } \mu_B \quad \text{co-efficient of friction on ice road. 3}$$

$$\text{now use } d = 18\text{m} \quad u^2 = 2gd\mu_B$$

$$u = \sqrt{2 \times 9.8 \times 18 \times 0.2} = 8.4 \text{ m/s} \quad (2 \text{ sig fig}) \text{ or } (\sim 30 \text{ km/h})$$

Method 2

deceleration

$$\text{Find max deceleration on ice, } a = \frac{-u^2}{2s} = -\frac{(16.67)^2}{2 \times 71} = -1.956 \text{ m/s}^2.$$

Find initial speed at this deceleration to stop in 18m. ①

$$u = \sqrt{v^2 - 2as} = \sqrt{-2as} \quad \text{use } 18\text{m} \quad \text{①}$$

$$= \sqrt{-2 \left(\frac{-u^2}{2s} \right) \times 18}$$

$$= \sqrt{(16.67)^2 \times \frac{18}{71}}$$

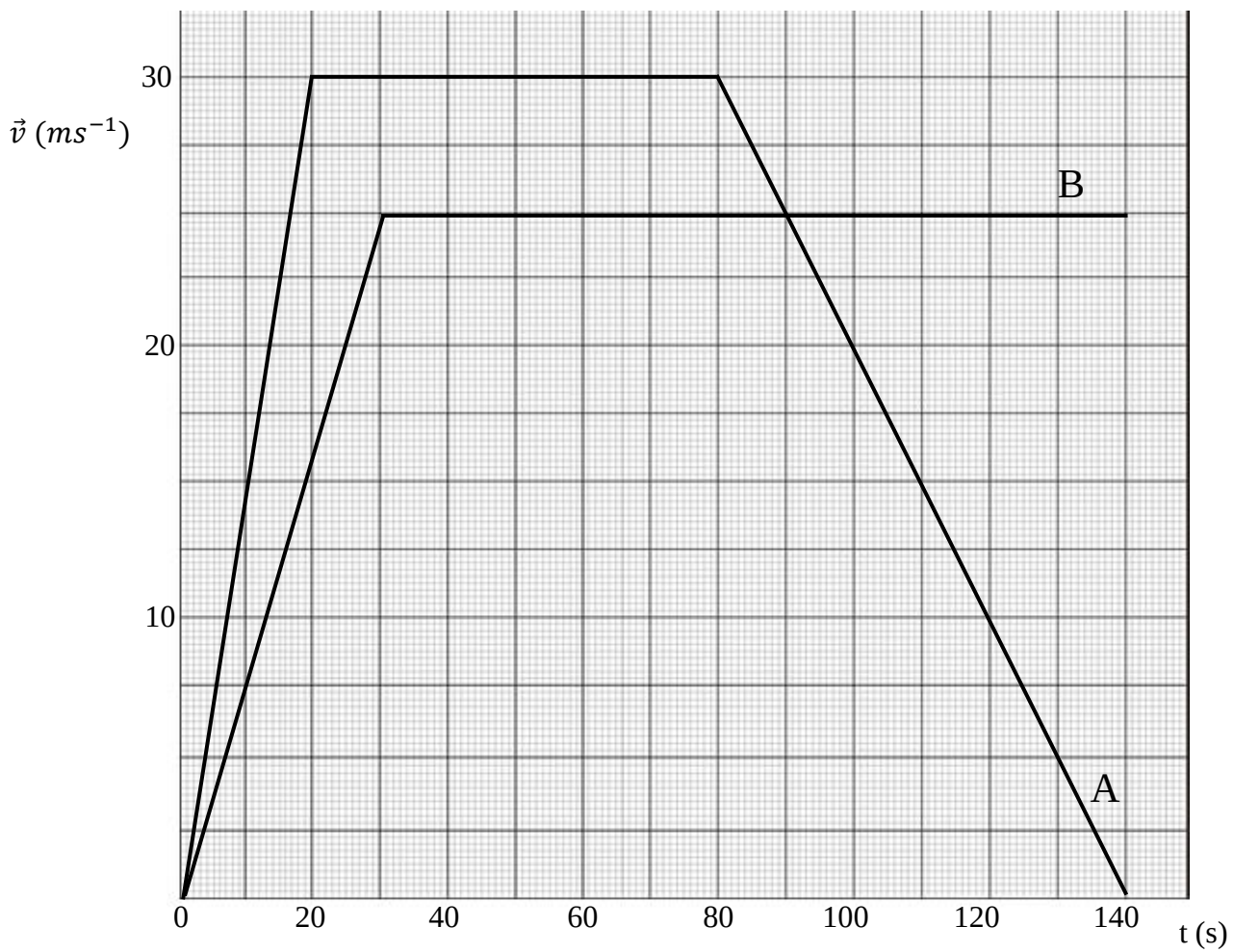
$$= 8.4 \text{ m/s.} \quad \text{① correct answer}$$

AAH - CRIB

Question 29 (6 marks)

Marks

The velocity versus time graphs below show two underground trains, A and B, travelling from the same station on parallel, straight tracks. At time $t = 0$ s, both trains are alongside each other at rest.



Question 29 continued on next page.

Question 29 continued.

Marks

Using the graphs, or otherwise, determine:

- (a) the maximum speed of train B.

From the Graph: = **25 ms⁻¹** **(1 mk)**

- (b) the magnitude of the acceleration of train A between 80 and 140 s.

= Gradient of Graph A from 80 – 140 s (1 mk for valid substitution)

= $30/60 = \mathbf{0.5\ ms^{-2}}$ **(2 mks)**

- (c) the time at which train A and train B are alongside each other again.

Start from $t = 80\text{ s}$: at this time $r_A = 2100\text{ m}$ and $r_B = 1625\text{ m}$
so B is 475 m behind A. - (1 mk)

$$r_B = 25.t \qquad r_A = 30.t - t^2/4$$

they meet when $r_B = r_A + 475$

i.e. when $25t = 30t - t^2/4 + 475$ - (2 mks)

solution: $t = 54.7\text{ s}$ - **(3 mks)**

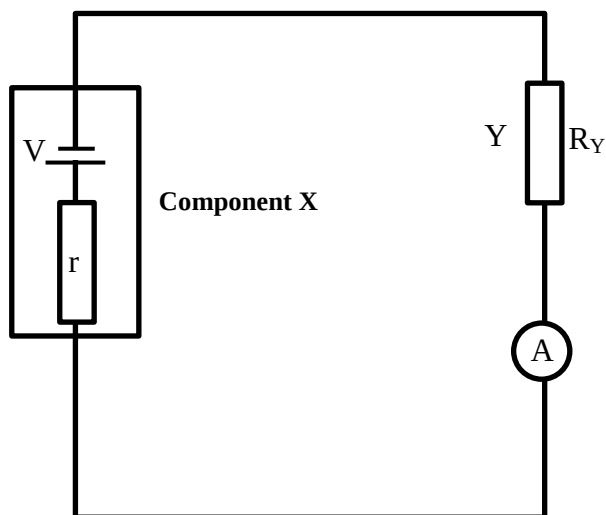
so, **$t_{\text{TOTAL}} = 134.7\text{ s}$**

NB: (1 mk) for any valid measurement of distance at any *stated* time

Question 30 (4 marks)

Marks

A physics student sets up the following circuit in an attempt to determine the electrical characteristics of component X.



Component X consists of a battery of potential difference, V and a resistor, r , in series.

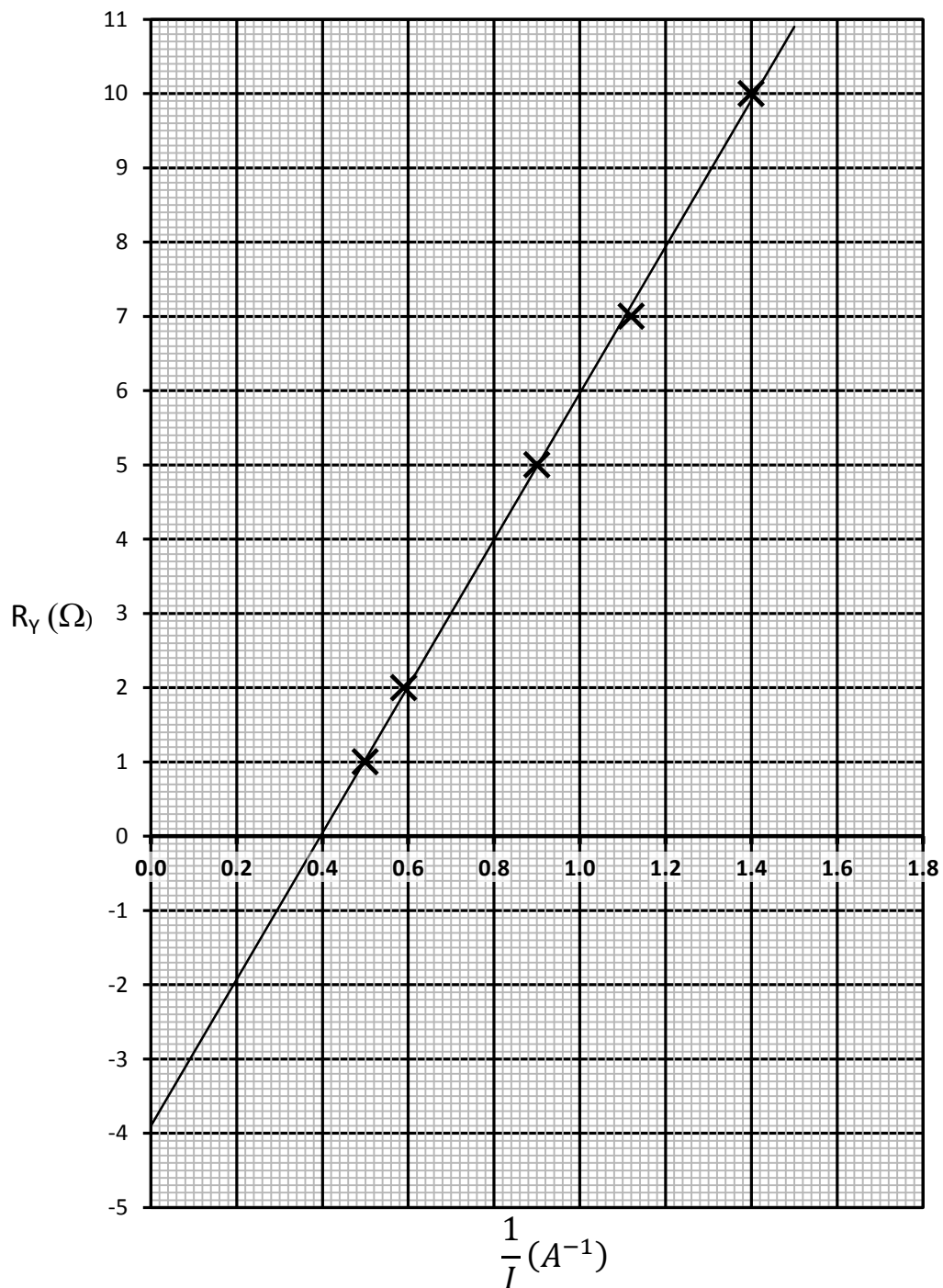
Various resistors, R_Y , are placed in the circuit at Y and the current flowing through the ammeter is recorded each time.

Question 30 continued on next page.

Question 30 continued.

Marks

The following graph is a plot of R_Y (on the y-axis) versus $\frac{1}{I}$ using the experimental results obtained.



Question 30 continued on next page.

Question 30 continued.

Marks

- (a) Use the graph to find the value of the resistor, r , within component X.

NB circuit analysis gives:

$$V = I (R_Y = r) \quad \text{so} \quad R_Y = V/I - r$$

r is the negative intercept of the graph (1 mk)

$$= \underline{3.9 \, \Omega} \text{ (accept } 4 \, \Omega) \quad \underline{(2 \text{ mks})}$$

- (b) Use the graph to find the value of the potential difference, V , of the battery within component X.

From above, $V = \text{Gradient of graph}$

$$= (10 - 0)/(1.4 - 0.4) \quad (1 \text{ mk})$$

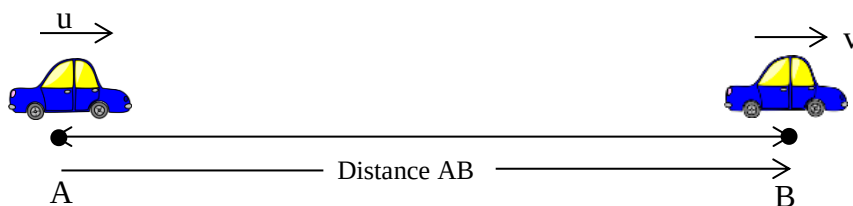
$$= \underline{10 \, V} \quad \underline{(2 \text{ mks})}$$

NB: carry through marks awarded only if the method was clear (e.g. if it was obvious where the numerical values used came from).

Question 31 (10 marks)

Marks

In an engine test, a car accelerates uniformly between two points (from the same starting velocity) on a straight road, and its final velocity is recorded.



The data obtained is included in the table on the following page.

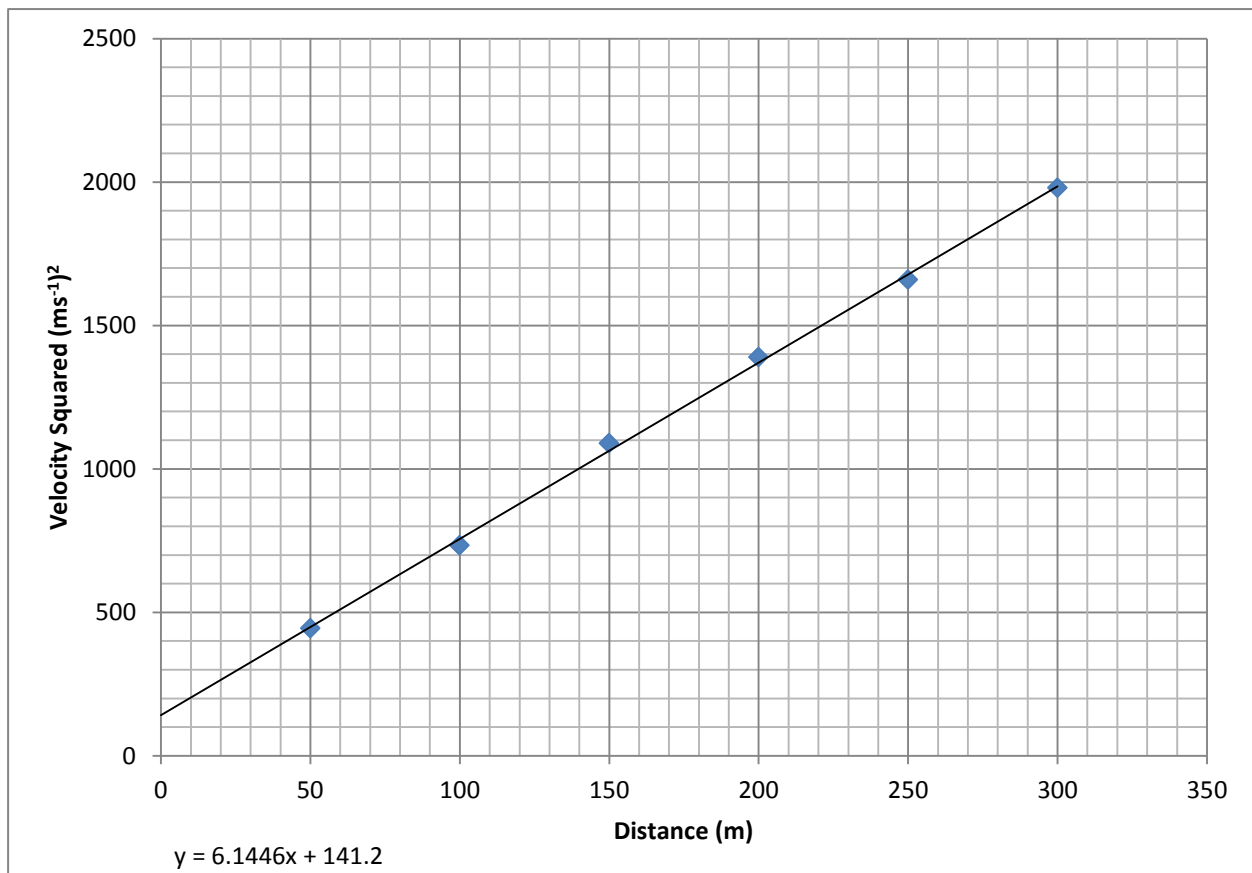
Distance AB (m)	Final Velocity, v (ms^{-1})			Average v (ms^{-1})*	Average v^2 (m^2s^{-2})*
	1	2	3		
50	20.6	21.4	21.2	21.1	445
100	27.2	27.2	27.0	27.1	734
150	32.7	33.4	32.8	33.0	1090
200	37.3	37.5	37.1	37.3	1390
250	40.4	40.9	40.8	40.7	1660*
300	44.2	44.6	44.8	44.5	1980

(* to 3 significant figures, with v^2 based on v to 3 significant figures)

*** sig figs not penalised here!**

- (a) Complete the table above by adding in the missing values.

(b) Plot a graph of velocity squared (on the y-axis) against distance.



- Points Correctly Plotted
- Axes Labelled
- Correct Units
- Appropriate Scale
- Line of Best Fit (which does NOT go through origin)
- Clarity

Minus 1 mk for any of the above missing

Question 31 continued.

Marks

(c) Using your graph, or otherwise, determine:

(i) the initial velocity of the car.

Intercept of the graph = $u^2 = 140$

So **$u = 11.8 \text{ ms}^{-1}$**

(accepted 11.0 – 13.5 for **2 mks**, with a wider range for 1 mk)

(ii) the acceleration of the car.

$v^2 = u^2 = 2ar$, so gradient of graph = $2a$

so, $2a = 6.1$

$a = 3 \text{ ms}^{-2}$

(I accepted 2.7 – 3.3 for **2 mks** – but was much more generous if the mark had already been lost in part (i))

NB: carry through mark awarded for part (ii) where possible, as long as the method was clear